Action Differences, Fluctuations and Operational Bounds

Control results for a research programme on mechanical gaps

navstokgap research working notes

6 September 2026 — working draft

Abstract

For a particle launched perpendicular to a constant force, we derive the exact chord–curve action difference $\Delta S = F^2T^3/(24m) = T\delta E/12$ and relate it to the geometric error area. Positive fixed-endpoint action differences accumulate at zero; a Taylor argument extends this result to nondegenerate variations. For constant force, the Dirichlet Jacobi operator has a positive lowest eigenvalue, separating normalized fluctuation cost from action amplitude. The free quantum kernel converges to a delta with a second-derivative short-time correction. Finally, within a specified quantum two-arm measurement model, we derive an exact action-resolution threshold for N independent copies and target success probability p. For fixed $1/2 < p < 1$, the threshold decreases as $N^{-1/2}$. These results connect trajectory geometry, fluctuation spectra and operational resolution in a tractable setting for studying gap formation.

1 Action values, fluctuations and resolution

The motivating question is whether limitations on Newtonian trajectories can explain a nonzero action scale, and whether a useful toy model can isolate the formation of a gap. We write $h = 2\pi\hbar$ for Planck's constants, η for a path variation, and J for a second-variation operator. An additional field χ would enter through a specified extended action $S[q, \chi]$.

For a path class $\mathcal{P}$ and reference q_*, one possible action-gap question is

$$g_S = \inf\{|S[q] - S[q_*]| : q \in \mathcal{P},\ S[q] \neq S[q_*]\} > 0.$$

Assume the set is nonempty and specify the path class, topology, endpoints and duration. The corresponding Hamiltonian gap question is $\operatorname{spec}(H) \cap (E_0, E_0 + \Delta) = \varnothing$ for some $\Delta > 0$. Here E_0 is the ground energy of a specified self-adjoint Hamiltonian. The path-space infimum and spectral interval give two precise meanings of a gap.

Newton's geometry motivates the area comparison [7, 8]. NATP00385 [6] supplies Newton's later account of analytic discovery and synthetic presentation. The following calculation expresses the geometric comparison through Hamilton's action.

2 Constant force: geometry and action

Fix $m, F, T, v_0 > 0$, set $V(y) = -Fy$, and impose $x(0) = y(0) = \dot{y}(0) = 0$, $\dot{x}(0) = v_0$. Newton's equations give

$$x_*(t) = v_0 t, \qquad y_*(t) = \frac{Ft^2}{2m}, \qquad 0 \le t \le T.$$

The kinetic-energy gain is $\delta E = F^2T^2/(2m) = -\Delta V$; total mechanical energy is conserved. The triangle between initial tangent, endpoint vertical and chord has area $v_0FT^3/(4m)$. The chord–curve lens has one third of this:

$$\mathcal{A}_{\text{lens}} = \frac{v_0 F T^3}{12m}.$$

These are frame-dependent areas in configuration space. The factor relating the lens area to action will follow from a matched-endpoint variation.

Hold $x = x_*$ fixed and consider the same-endpoint paths $y = y_* + \eta$ with $\eta \in H_0^1(0,T)$, the Sobolev space with square-integrable weak derivative and zero endpoint trace. Define

$$S[x,y] = \int_0^T \left[\frac{m}{2}(\dot{x}^2 + \dot{y}^2) + Fy\right] \mathrm{d}t.$$

The linear variation vanishes by integration by parts and $m\ddot{y}_* = F$:

$$S[x_*, y_* + \eta] - S[x_*, y_*] = \frac{m}{2}\int_0^T \dot{\eta}^2 \,\mathrm{d}t. \tag{1}$$

The straight chord $y_{\text{ch}} = FTt/(2m)$ is an admissible off-shell comparison path. Substitution into (1) gives

$$\Delta S_{\text{ch},*} = \frac{F^2T^3}{24m} = \frac{F}{2v_0}\mathcal{A}_{\text{lens}} = \frac{T\delta E}{12}. \tag{2}$$

Adding the same total time derivative $\mathrm{d}G(q,t)/\mathrm{d}t$ to both Lagrangians leaves this difference unchanged because configuration endpoints and duration coincide. A velocity-dependent endpoint function requires additional endpoint data.

Proposition 1 (Zero infimum of action differences)**.** *For this fixed-endpoint path class the infimum of positive action differences from the classical path is zero, although the classical path is the unique minimizer within the class with fixed horizontal motion.*

Proof. If $\eta \neq 0$ in H_0^1, its derivative cannot vanish almost everywhere, so (1) is positive. For $\eta_a(t) = at(T-t)$ it equals

$$\Delta S(a) = \frac{ma^2T^3}{6}.$$

Any prescribed positive bound is violated by taking a sufficiently small nonzero a. The zero variation alone gives equality in (1). □

Proposition 2 (Local obstruction along a variation)**.** *Let q_a be admissible for all a in an open interval about zero, and suppose $f(a) = S[q_a]$ is twice continuously differentiable there, with $f'(0) = 0$ and $f''(0) = b \neq 0$. Then arbitrarily small nonzero absolute action differences $|S[q_a] - S[q_0]|$ occur. If $b > 0$, these differences can be taken positive without absolute values.*

Proof. Taylor's formula gives $f(a) - f(0) = ba^2/2 + o(a^2)$. For all sufficiently small nonzero a, its magnitude lies between $|b|a^2/4$ and $3|b|a^2/4$ and its sign is the sign of b. It is nonzero and tends to zero. □

Locality here is initially in the parameter a. To conclude that these actions occur in every path-space neighbourhood of q_0, additionally require $q_a \to q_0$ in the specified path topology. The polynomial family above converges in H^1 (and in C^1).

The admissible small-amplitude family is the mechanism behind both propositions. Discrete path classes, topological sectors and operational equivalence relations offer concrete ways to change this admissibility question. Kepler collisions pose a separate IVP question about continuation through a singular force.

3 Jacobi fields and normalized fluctuation gaps

For $S[q] = \int_0^T (m\dot{q}^2/2 - V(q))\mathrm{d}t$ about a smooth classical solution, the second variation is

$$Q[\eta] = \int_0^T \left[m\dot{\eta}^2 - V''(q_*)\eta^2\right] \mathrm{d}t, \qquad J = -m\frac{\mathrm{d}^2}{\mathrm{d}t^2} - V''(q_*).$$

For smooth bounded $V''(q_*(t))$ take the Dirichlet operator domain $H^2(0,T) \cap H_0^1(0,T)$ in $L^2(0,T)$. A variation through classical solutions satisfies the Jacobi equation $J\eta = 0$. The full variational path class also contains off-shell fluctuations. For constant force a Jacobi field is $A+Bt$, so fixed zero endpoints force it to vanish. Equation (1) nevertheless admits nonzero fluctuations.

The second-variation framework is classical [1, §8.2]. For constant force the Dirichlet eigenfunctions are $\sin(n\pi t/T)$ with eigenvalues $\lambda_n = m(n\pi/T)^2$, $n \geq 1$. Thus

$$Q[\eta] \geq \frac{m\pi^2}{T^2}\|\eta\|_{L^2}^2.$$

This is coercivity relative to the L^2 norm: rescaling η rescales Q quadratically. The eigenvalue λ_n has units mass/time^2, while $\|\eta\|_{L^2}^2$ has units length^2 time, producing action.

For the oscillator $V = m\omega^2q^2/2$, the corresponding eigenvalues become $m[(n\pi/T)^2 - \omega^2]$. Zero modes occur at $T = n\pi/\omega$; positivity of all modes fails after the first such time. This explicit calculation is a useful comparison with the quantum oscillator Hamiltonian. The companion working draft *A Gap Laboratory: Oscillators, Circles and Winding Sectors* derives both spectra, their domains and their distinct gap-closing limits.

4 What distribution is the oscillatory kernel?

The quantum path-phase input is $\exp(iS/\hbar)$, with time-integrated action, as in the Feynman formulation [3, 4]. To study its distributional meaning, specify a kernel, normalization and limiting parameter. The free one-dimensional Schrödinger kernel provides an exact example [9, §7.3, (7.30)]. The expansion below is its Fourier consequence. For $\hbar, m, \tau > 0$ let

$$K_\tau(z) = \left(\frac{m}{2\pi i\hbar\tau}\right)^{1/2} \exp\left(\frac{imz^2}{2\hbar\tau}\right),$$

with the usual branch $1/\sqrt{i} = e^{-i\pi/4}$ and oscillatory/tempered-distribution interpretation. Use Fourier transform $\widehat{f}(k) = \int e^{-ikz}f(z)\mathrm{d}z$. Equivalently this kernel is defined by

$$\widehat{K_\tau}(k) = \exp\left(-\frac{i\hbar\tau k^2}{2m}\right).$$

For each Schwartz test function, dominated convergence in Fourier space implies

$$K_\tau \longrightarrow \delta \quad (\tau \downarrow 0,\ \hbar > 0 \text{ fixed}). \tag{3}$$

Taylor expansion of the multiplier, using $|e^{-iu} - 1 + iu| \leq u^2/2$ for real u, further gives

$$K_\tau = \delta + \frac{i\hbar\tau}{2m}\delta'' + O(\tau^2) \quad \text{in the sense of pairing with each fixed Schwartz test function.}$$

The remainder estimate follows by integrating k^4 times the absolute value of the transformed test function. The even free kernel gives δ at leading order and δ'' at first order in time. Differentiating the kernel with respect to z instead gives a family converging to δ'. A Fourier constraint integral provides another construction: $\int e^{i\lambda z/\hbar}\mathrm{d}\lambda/(2\pi\hbar) = \delta(z)$. Substitution of a nonlinear constraint requires further regularity and Jacobian conditions.

The limit above is short time at fixed $\hbar$. A semiclassical analysis instead varies $\hbar$; each limit has its own test functions and convergence statement.

5 A conditional operational action threshold

Assume the quantum state/measurement framework, a specified positive $\hbar$, and a coherently controlled two-arm system realizing a relative phase $\phi = \Delta S/\hbar$. The relative arm phase encodes two known hypotheses with equal priors:

$$|\psi_0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |\psi_\phi\rangle = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}.$$

Allow $N \geq 1$ independent identical copies and any joint binary quantum measurement. These assumptions define the resource budget and the hypothesis test.

The following is a pure-state specialization of the Holevo–Helstrom theorem [10, Theorem 3.4], followed by the specified action-phase encoding.

Proposition 3 (Finite-copy bound)**.** *The optimal equal-prior success probability is*

$$P_N(\phi) = \frac{1}{2}\left(1 + \sqrt{1 - \cos^{2N}(\phi/2)}\right). \tag{4}$$

For $1/2 < p \leq 1$, in the first phase lobe $|\phi| \leq \pi$, success at least p is possible if and only if

$$|\Delta S| \geq d_{N,p} := 2\hbar \arccos\left([4p(1-p)]^{1/(2N)}\right). \tag{5}$$

For each fixed $1/2 < p < 1$, $d_{N,p} \to 0$ as $N \to \infty$. For $p = 1$, the finite-copy threshold instead remains $\pi\hbar$.

Proof. Two normalized pure states with overlap magnitude r have density-matrix difference D with eigenvalues $\pm\sqrt{1-r^2}$ on their span (and zero outside). Indeed $\operatorname{tr} D = 0$ and $\operatorname{tr} D^2 = 2(1-r^2)$. A binary measurement effect $0 \leq E \leq I$ gives success $1/2 + \operatorname{tr}(ED)/2$, maximized by projecting onto the positive eigenspace. The maximum is $(1+\sqrt{1-r^2})/2$. Here a single-copy overlap has magnitude $|\cos(\phi/2)|$, and tensor products raise it to the Nth power, proving (4). Within the first lobe cosine is nonnegative and monotone in $|\phi|$, so solving $P_N \geq p$ gives (5). For $0 < 4p(1-p) < 1$ its $1/(2N)$ power tends to one. At $p = 1$ it is zero for every finite N, giving the stated exception. □

For example $N = 1$, $p = 3/4$ gives $d_{1,3/4} = \pi\hbar/3$. For fixed $1/2 < p < 1$ the asymptotic threshold is

$$d_{N,p} \sim 2\hbar\sqrt{\frac{-\log(4p(1-p))}{N}}.$$

The threshold is positive at fixed resources and accuracy. Its dependence on N quantifies the improvement in resolution from repeated preparations. Phase periodicity explains the first-lobe restriction: at $\phi = 2\pi n$ the hypotheses coincide again.

If an apparatus realizes the particular relative action (2), with all control and recombination phases included, its first-lobe threshold would be

$$T \geq \left(\frac{24 m d_{N,p}}{F^2}\right)^{1/3}.$$

The upper first-lobe condition $F^2T^3/(24m\hbar) \leq \pi$ must be retained. Realizing the off-shell chord as an experimental arm requires a control force; the displayed substitution applies when the full apparatus has the specified relative action.

6 Finite propagation and the field-theory companion

For the free relativistic Lagrangian $L = -mc^2\sqrt{1-\dot{q}^2/c^2}$, consider the fixed-endpoint family $q_a = at(T-t)$, with both endpoints zero. If $|a|T < c$, the path has a strict speed margin. Relative to the rest path its action excess is positive for $a \neq 0$ and

$$\Delta S(a) = mc^2 \int_0^T \left(1 - \sqrt{1 - a^2(T-2t)^2/c^2}\right) \mathrm{d}t = \frac{ma^2T^3}{6} + O(a^4) \longrightarrow 0.$$

Thus positive action differences accumulate at zero within the strict speed margin, just as in the Newtonian example. This gives a kinematic comparison at finite c.

Navier–Stokes and Yang–Mills remain complementary comparisons, with their official targets kept intact [2, 5]. A diffusion or Dirichlet coercivity bound can depend on time, domain and normalization; a physical quantum mass gap concerns a different operator and limiting problem. Any future bridge must identify the spaces, physical versus auxiliary time, estimates and limits it preserves. The fluid target is regularity of evolution; the field-theory target includes construction of the physical quantum theory and its vacuum gap.

7 Scope, reproduction and next experiments

The results concern the stated mechanical path classes and quantum measurement model. Hilbert states, the Born rule and positive $\hbar$ are physical inputs to Section 5. Deriving these premises belongs to the reconstruction branch of the companion programme. The proofs received internal review; novelty assessment and external review are publication-stage tasks.

The repository's `make check` verifies the exact polynomial action identities, the oscillator-mode residual, the Fourier expansion coefficient and elementary discrimination cross-checks. `make papers` rebuilds this note and the companion programme PDF. The formalisation queue begins with the small-amplitude polynomial lemma.

The companion *Time Refinement and an Action Scale* develops the free Gaussian blocking test. Next steps audit the regulator proposal and quantum reconstruction axioms, alongside central-potential models and the historical corpus inventory, with sequential source audits. A successful extension must identify which premise excludes the classical small-amplitude families, or else choose an operational or spectral gap instead and state the change of target. Deriving a theory's form, forcing a nonzero universal scale and determining its measured numerical value are separate tasks.

References

[1] Riccardo Cristoferi. Calculus of variations: Lecture notes, 2016. Dated 9 May 2016; Section 8.2, printed pp. 79–80.

[2] Charles L. Fefferman. Existence and smoothness of the Navier–Stokes equation. Clay Mathematics Institute, official Millennium problem description. Local source companion records statement alternatives and reading coverage.

[3] Richard P. Feynman. Space-time approach to non-relativistic quantum mechanics. *Reviews of Modern Physics*, 20:367–387, 1948. Publisher metadata and abstract checked; no full-paper proof audit.

[4] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. The Feynman lectures on physics, volume ii, chapter 19: The principle of least action. California Institute of Technology, online edition. Section 19–1, quantum discussion following Figure 19–11; consulted 2026-09-05.

[5] Arthur Jaffe and Edward Witten. Quantum Yang–Mills theory. Clay Mathematics Institute, official Millennium problem description. Local source companion records existence and mass-gap requirements.

[6] Isaac Newton. Unarranged fragments, mostly relating to the dispute with Leibniz. The Newton Project, text NATP00385. MS Add. 3968, ff. 594r–619v, Cambridge University Library; electronic transcription completed by Robert Ralley in 2020; consulted 2026-09-05. Manuscript collection has no single composition date assigned here.

[7] Isaac Newton. *Newton's Principia: The Mathematical Principles of Natural Philosophy*. Daniel Adee, New York, 1846. Andrew Motte translation; N. W. Chittenden edition. Definitions, Laws, Book I Sections I–II; local source companion records reading coverage.

[8] Isaac Newton. The mathematical principles of natural philosophy, book i, section i, 2002. Andrew Motte's 1729 translation, edited by David R. Wilkins; local 11-page excerpt.

[9] Gerald Teschl. *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, volume 99 of *Graduate Studies in Mathematics*. American Mathematical Society, 1 edition, 2009. Author-hosted first edition, version 12 February 2009; selected Sections 7.2–7.3 and 8.3.

[10] John Watrous. *The Theory of Quantum Information*. Cambridge University Press, 2018. Author-hosted manuscript, Chapter 3; Theorem 3.4, pp. 128–129, and equation (3.20).

Time Refinement and an Action Scale

navstokgap research working notes

7 September 2026

Abstract

Free-particle kernels give an exact test of refinement-based proposals for an action scale. Integrating over an inserted position preserves a Gaussian family; time homogeneity makes its variance linear in duration. The proportionality constant remains a model parameter. Fixed-endpoint bridges concentrate on the Newtonian straight path as that parameter vanishes. We use these standard calculations to isolate the physical selection question in the proposed connection between geometric subdivision, renormalisation and quantum mechanics.

1 A source-driven experiment

The experiment is to refine a time interval, integrate out the inserted positions, and track the remaining parameters. Rivero's cone discussion [5] motivates retaining rescaled data as sections approach; his path-integral proposal [4] motivates a finite blocking map. Brouder's rooted-tree correspondence [1] supplies an algebraic route to nonlinear composition, and the tangent-groupoid construction [2] supplies a separate quantum-completion reading.

This note works on the free line with mass $m > 0$. Positions have units of length, $s, t, T > 0$ are durations, and κ has units of action. An identification $\kappa = \hbar = h/(2\pi)$ is a physical interpretation of a selected oscillatory model. The present calculations classify a free family; the selection of its physical member is the open reconstruction question.

2 Eliminating an inserted position

The normalized positive kernel is

$$G_t^\kappa(x) = \sqrt{\frac{m}{2\pi\kappa t}} \exp\left(-\frac{mx^2}{2\kappa t}\right), \qquad \kappa > 0. \tag{1}$$

It is a probability density on $\mathbb{R}$ with variance $\kappa t/m$; the standard Brownian normalization is given in [3, §4.1, (23)–(25)]. Its exponent is minus the free straight-segment action divided by κ. Here positivity defines a probability model; the oscillatory model enters in Section 4.

Proposition 1 (Exact blocking)**.** *For $s, t, \kappa > 0$ and $x, z \in \mathbb{R}$,*

$$\int_{\mathbb{R}} G_t^\kappa(z-y) G_s^\kappa(y-x)\, dy = G_{s+t}^\kappa(z-x).$$

Proof. Set $\mu = (tx + sz)/(s+t)$. Completing the square gives

$$\frac{(y-x)^2}{s} + \frac{(z-y)^2}{t} = \frac{(z-x)^2}{s+t} + \frac{s+t}{st}(y-\mu)^2.$$

The remaining Gaussian integral is $\sqrt{2\pi\kappa st/[m(s+t)]}$. Multiplication by the two prefactors in (1) gives exactly the prefactor at duration $s+t$. $\square$

Consequently, for any finite partition $0 = t_0 < \cdots < t_n = T$, integrating the product of the n kernels over its $n-1$ internal positions gives $G_T^\kappa(z-x)$. The action parameter is unchanged at every finite refinement; this identity requires no continuum path measure.

Proposition 2 (The Gaussian parameter left by composition)**.** *Let $(\mu_t)_{t\geq 0}$ be a weakly continuous convolution semigroup of centered Gaussian probability measures on $\mathbb{R}$, allowing a degenerate Gaussian, with $\mu_0 = \delta_0$. Then μ_t has variance $w(t) = at$ for one $a \geq 0$. Conversely every $a \geq 0$ defines such a family.*

Proof. Characteristic functions are $\widehat{\mu}_t(k) = \exp[-w(t)k^2/2]$. Composition implies $w(s+t) = w(s) + w(t)$. Weak continuity, evaluated at $k = 1$, makes w continuous. Additivity first yields $w(r) = rw(1)$ for nonnegative rational r, and continuity extends this to every real $t \geq 0$. Nonnegative variance gives $a = w(1) \geq 0$. These characteristic functions verify the converse. □

At fixed mass set $\kappa = ma$; since $[a] = \text{length}^2/\text{time}$, $[\kappa] = \text{action}$. All $\kappa > 0$ satisfy the same composition law, and $a = 0$ supplies the identity convolution family. Positive-width densities select $a > 0$ as an additional premise. Even that subclass has $\inf\{\kappa : \kappa > 0\} = 0$: positivity of each member and a uniform lower bound over a class are distinct questions.

3 A bridge to the Newtonian path

Conditioning on endpoints gives an exact trajectory comparison. The standard bridge construction is given in [3, §3.4]. For $0 < s < T$, define the normalized intermediate-position density

$$\rho_s^\kappa(y \mid x, z) = \frac{G_{T-s}^\kappa(z-y)G_s^\kappa(y-x)}{G_T^\kappa(z-x)}.$$

Proposition 1 makes its integral one. The same square completion gives mean and variance

$$q_{\text{cl}}(s) = x + \frac{s}{T}(z-x), \qquad \sigma_s^2 = \frac{\kappa s(T-s)}{mT}. \tag{2}$$

The mean is the unique free Newtonian trajectory with these endpoints. For any displacement tolerance $r > 0$, Chebyshev's inequality gives

$$\Pr\{|Y_s - q_{\text{cl}}(s)| \geq r\} \leq \frac{\kappa s(T-s)}{mTr^2} \longrightarrow 0.$$

For any fixed finite set of internal times, the product-kernel bridge has these marginals, and a union bound proves joint concentration on the sampled straight path. This is a finite-dimensional classical-path limit in a positive-measure model. Quantum phase-space correspondence is a further limiting problem.

The zero-variance convolution family describes a stationary position. General free classical motion instead has the phase-space transition measure

$$P_t((q,p), dq'\,dp') = \delta_{q+pt/m}(dq')\,\delta_p(dp'),$$

whose composition follows by substitution. This keeps the deterministic comparison on its appropriate state space.

4 The oscillatory family and its coefficient

For each $\kappa > 0$, the free oscillatory kernel is

$$K_t^\kappa(x) = \sqrt{\frac{m}{2\pi i\kappa t}} \exp\left(\frac{imx^2}{2\kappa t}\right), \qquad t > 0.$$

Use the unitary transform $\mathcal{F}f(k) = (2\pi)^{-1/2} \int_{\mathbb{R}} e^{-ikx} f(x)\, dx$. The kernel's precise meaning is the tempered-distribution kernel of the Fourier multiplier $\exp[-i\kappa t k^2/(2m)]$, with the square-root branch fixed by continuation from positive real Gaussian variance. On $L^2(\mathbb{R})$, these multipliers define a strongly continuous unitary group for real t: the group law follows by multiplication, unitarity by modulus one, and strong continuity by dominated convergence. Its Hamiltonian in the convention $i\kappa\partial_t\psi = H_\kappa\psi$ is

$$H_\kappa = -\frac{\kappa^2}{2m}\frac{d^2}{dx^2}, \qquad D(H_\kappa) = H^2(\mathbb{R}).$$

These are the usual free-particle formulas [6, §7.3]. Operator composition defines the oscillatory convolution, avoiding an unqualified interchange of conditionally convergent integrals.

Writing the positive family as $\exp[t\kappa\partial_x^2/(2m)]$ and continuing t to it yields this unitary family. That continuation is a specified model choice. Both constructions retain a free coefficient with action units.

5 The selection problem now exposed

The first task is to find a physical principle that selects a positive action scale and explains its universality under composition of interacting systems. Three parameters stay separate: temporal mesh size, the action scale κ, and any finite-speed parameter c^{-1}. The present free nonrelativistic model allows arbitrarily fine temporal meshes at each fixed κ.

The next source-guided experiment audits the two regulators in [4], starting with a finite-dimensional quadratic action and its exact normalization. A second task extracts the hypotheses and limiting topology of [2]. Together they test whether refinement consistency constructs a quantum completion, and which additional premise selects its physical action parameter. Interactions and finite propagation then become explicit changes of model.

References

[1] Christian Brouder. Runge–Kutta methods and renormalization. arXiv:hep-th/9904014v1, 1999.

[2] José F. Cariñena, Jesús Clemente-Gallardo, Eduardo Follana, José M. Gracia-Bondía, Alejandro Rivero, and Joseph C. Várilly. Connes' tangent groupoid and strict quantization. *Journal of Geometry and Physics*, 32:79–96, 1999. arXiv:math/9802102v3.

[3] Jim Pitman and Marc Yor. A guide to Brownian motion and related stochastic processes. arXiv:1802.09679v1, 2018.

[4] Alejandro Rivero. A short derivation of Feynman formula. arXiv:quant-ph/9803035v1, 1998.

[5] Alejandro Rivero. On the section of a cone. arXiv:math/9904021v1, 1999.

[6] Gerald Teschl. *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, volume 99 of *Graduate Studies in Mathematics*. American Mathematical Society, 1 edition, 2009. Author-hosted first edition, version 12 February 2009; selected Sections 7.2–7.3 and 8.3.

Classical Paths with a Finite Action Defect

An exact two-regulator experiment

navstokgap research working notes

7 September 2026

Abstract

A shrinking Gaussian bridge can converge uniformly to the free classical path while its excess kinetic action converges to a positive constant. With d internal positions and fluctuation action scale κ, that constant is controlled by $d\kappa$: the exact mean excess is $d\kappa/2$. We derive the partition Hessian, normalization and limiting law, then audit a two-regulator path-integral proposal. A bounded-speed oscillatory example isolates the role of derivative control. The calculation turns the search for an action gap into a concrete selection question: which physical premise fixes a positive surviving quantity across changes of resolution?

1 The object retained by a limit

The central object is the *action defect* $\lim_N(S[q_N] - S[q_{\rm cl}])$ for specified convergent paths and fixed endpoints. Uniform path convergence and convergence of kinetic action impose different demands on their derivatives. This distinction is familiar from oscillations in the calculus of variations [3, §§2.1,3.3]; here it gives an exactly soluble mechanics test.

Work on the free line during $[0, T]$, with $T > 0$, mass $m > 0$ and endpoints $x, z \in \mathbb{R}$. Positions have length units. The parameter $\kappa > 0$ has action units and defines a positive Gaussian path model. We separately examine a complex oscillatory integral with action parameter $\epsilon > 0$. The paths integrated over are off-shell comparison paths; the free solution is $q_{\rm cl}(t) = x + t(z - x)/T$. The results concern these specified models. For the proposed universal quantum scale, the physical selection principle and the justification of a phase rule remain the research obligations.

2 Exact normalization at any partition

Let $\pi = (0 = t_0 < \cdots < t_N = T)$, $N \geq 2$, $\tau_j = t_j - t_{j-1}$ and $d = N - 1$. The piecewise-linear path with nodes $q_0 = x, q_N = z$ has action

$$S_\pi(q) = \frac{m}{2}\sum_{j=1}^{N}\frac{(q_j - q_{j-1})^2}{\tau_j}.$$

Set $\eta_j = q_j - q_{\rm cl}(t_j)$, with $\eta_0 = \eta_N = 0$. Summation of the cross term gives

$$S_\pi = S_{\rm cl} + \tfrac{1}{2}\eta^T A_\pi \eta, \qquad S_{\rm cl} = \frac{m(z - x)^2}{2T}. \tag{1}$$

The $d \times d$ matrix has entries

$$(A_\pi)_{jj} = m(\tau_j^{-1} + \tau_{j+1}^{-1}), \qquad (A_\pi)_{j,j+1} = (A_\pi)_{j+1,j} = -m\tau_{j+1}^{-1},$$

and zeros elsewhere. Its units are mass/time, since its quadratic form acts on node displacements. The continuous Hessian with the $L^2(dt)$ inner product instead has eigenvalues with units mass/time squared.

Proposition 1 (Partition determinant and covariance)**.** *The matrix A_π is positive definite and*

$$\det A_\pi = \frac{m^d T}{\prod_{j=1}^N \tau_j}, \qquad (A_\pi^{-1})_{ij} = \frac{1}{m}\left(\min(t_i,t_j) - \frac{t_i t_j}{T}\right). \tag{2}$$

The normalized endpoint bridge density with respect to $d\eta_1 \cdots d\eta_d$ is

$$\rho_\pi^\kappa(\eta) = \frac{\sqrt{\det A_\pi}}{(2\pi\kappa)^{d/2}} \exp\left(-\frac{\eta^T A_\pi \eta}{2\kappa}\right), \qquad \operatorname{Cov}(\eta) = \kappa A_\pi^{-1}. \tag{3}$$

Proof. The quadratic form is $m\sum_j(\eta_j - \eta_{j-1})^2/\tau_j$; its vanishing forces every node to zero. For the leading $k \times k$ minor, the tridiagonal determinant recurrence gives $D_k = m^k(\tau_1 + \cdots + \tau_{k+1})/(\tau_1 \cdots \tau_{k+1})$. This holds at $k = 0, 1$, and substitution into $D_k = m(\tau_k^{-1} + \tau_{k+1}^{-1})D_{k-1} - m^2\tau_k^{-2}D_{k-2}$ proves it inductively. For each column of the proposed inverse, the piecewise-linear Green function vanishes at the endpoints, has constant slope on either side of its node, and a slope decrease of $1/m$ there. Multiplication by A_π therefore gives the corresponding coordinate vector. Diagonalizing the positive matrix then gives the Gaussian integral and covariance in (3). □

There are N transition densities and $d = N - 1$ integrated positions:

$$\left(\prod_{j=1}^N \sqrt{\frac{m}{2\pi\kappa\tau_j}}\right) \int_{\mathbb{R}^d} e^{-S_\pi(q)/\kappa}\, dq = \sqrt{\frac{m}{2\pi\kappa T}} e^{-S_{\mathrm{cl}}/\kappa}. \tag{4}$$

Thus the determinant restores exactly the free transition kernel at duration T. Dividing the product by that kernel yields (3). This is the Brownian transition/bridge construction [2, §§3.4,4.1], with variance parameter κ/m.

3 A classical path limit with residual action

The number of modes and their individual scale enter the action through their product. This gives the promised two-regulator calculation.

Proposition 2 (Exact defect law and joint limit)**.** *Under* (3)*, the excess $\Delta S_\pi = S_\pi - S_{\mathrm{cl}}$ satisfies*

$$\frac{2\Delta S_\pi}{\kappa} \sim \chi_d^2, \qquad \mathbb{E}\Delta S_\pi = \frac{d\kappa}{2}, \qquad \operatorname{Var}(\Delta S_\pi) = \frac{d\kappa^2}{2}. \tag{5}$$

Let $N \to \infty$, $d = N - 1$, and choose partitions with maximum interval tending to zero. If $\kappa_N \to 0$ and $d\kappa_N \to \ell \in [0,\infty)$, their polygonal bridges admit a coupling for which

$$q_N \longrightarrow q_{\mathrm{cl}} \quad \textit{uniformly almost surely,} \qquad \Delta S_{\pi_N} \longrightarrow \ell/2 \quad \textit{in } L^2.$$

The quantity ℓ has action units.

Proof. Whitening gives $Z = A_\pi^{1/2}\eta/\sqrt{\kappa}$ with independent standard normal coordinates. Equation (1) becomes $\Delta S_\pi = \kappa\sum_{j=1}^d Z_j^2/2$. The normal moments $\mathbb{E}Z_j^2 = 1$, $\mathbb{E}Z_j^4 = 3$ prove (5) and imply

$$\mathbb{E}(\Delta S_{\pi_N} - \ell/2)^2 = d\kappa_N^2/2 + (d\kappa_N - \ell)^2/4 \longrightarrow 0.$$

For the path limit, take one continuous standard Brownian bridge $B_t^T = B_t-(t/T)B_T$ on $[0,T]$ and let I_π denote linear interpolation at the partition nodes. Its covariance is $\min(s,t)-st/T$ [2, p. 6, (4)–(5)]. Consequently $q_N = q_{\mathrm{cl}} + \sqrt{\kappa_N/m}\, I_{\pi_N} B^T$ has the required law, and

$$\|q_N - q_{\mathrm{cl}}\|_\infty \le \sqrt{\kappa_N/m}\, \|B^T\|_\infty \longrightarrow 0$$

almost surely by continuity on a compact interval. The action moment identity holds under this coupling as well. □

For equal steps $\tau = T/N$ and $\kappa_N = b\tau$, where $b > 0$ has energy units, $\ell = bT$ and the residual action is $bT/2$. More generally, $d\kappa_N \to 0$ gives zero residual action, whereas $d\kappa_N \to \infty$ gives divergence in probability: the relative variance is $2/d$. At fixed $d, \kappa > 0$, the support of ΔS_π is $[0,\infty)$. Thus the positive residual is a selected limiting value, rather than a hard lower bound on the positive actions of admissible finite paths.

4 A bounded-speed control example

A speed bound alone still permits action defects. For the Newtonian action on $H^1([0,T])$ with fixed endpoints, choose $c > |z-x|/T$ and $0 < v_* < c - |z-x|/T$. The smooth off-shell paths

$$q_n(t) = q_{\mathrm{cl}}(t) + \frac{v_* T}{2\pi n}\sin(2\pi n t/T) \tag{6}$$

converge uniformly to q_{cl} and satisfy $|\dot q_n| < c$, while

$$S[q_n] - S[q_{\mathrm{cl}}] = \frac{m v_*^2 T}{4} > 0. \tag{7}$$

Indeed the cross term integrates to zero and $\int_0^T \cos^2(2\pi n t/T)dt = T/2$. Their acceleration amplitude grows as $2\pi n v_*/T$. This isolates derivative and force control as the next dynamical test. Here c is an imposed kinematic bound on the Newtonian model; a relativistic Lagrangian is a separate choice.

The example is an elementary oscillation mechanism of the kind organized by Young measures [3, §3.3]. Its coefficient varies continuously with v_*. Strong convergence of velocities in L^2 would instead make this quadratic action converge to the action of the limiting path.

5 Quadratic stationary phase and critical-point weights

For one nondegenerate quadratic stationary point, a normalized oscillatory amplitude gives a precise version of Rivero's finite-dimensional construction [4, p. 1, (2)–(4)]. Let $F(q) = F_* + (q-q_*)^T A (q-q_*)/2$ with A real symmetric positive definite on $\mathbb{R}^d$, and $O \in \mathcal{S}(\mathbb{R}^d)$. Here F has action units and A has action/length squared units. Define

$$I_\epsilon[O] = (2\pi\epsilon)^{-d/2} \int_{\mathbb{R}^d} e^{iF(q)/\epsilon} O(q)\, dq.$$

Use $\widehat{O}(k) = (2\pi)^{-d/2}\int e^{-ik\cdot q} O(q)\, dq$. The Gaussian Fourier identity [1, p. 464, (14.7)] gives

$$\begin{aligned} e^{-iF_*/\epsilon} I_\epsilon[O] &= \frac{e^{i\pi d/4}}{\sqrt{\det A}} (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{ik\cdot q_*} e^{-i\epsilon k^T A^{-1} k/2} \widehat{O}(k)\, dk \\ &\longrightarrow \frac{e^{i\pi d/4}}{\sqrt{\det A}} O(q_*). \end{aligned}$$

One can obtain the identity by inserting Gaussian damping in the quadratic Fourier integral and removing it in tempered distributions. The limit follows from $\widehat{O} \in L^1$ and dominated convergence. In particular,

$$\lim_{\epsilon \downarrow 0} |I_\epsilon[O]|^2 = \frac{|O(q_*)|^2}{\det A} = \langle \delta^{(d)}(\nabla F), |O|^2 \rangle. \tag{8}$$

The last equality is the change of variables $p = A(q - q_*)$. Equation (8) is a scalar limit for each fixed test function O; the distribution on its right acts linearly on $|O|^2$. The amplitude carries a square-root determinant weight and a removable global phase; its squared modulus gives the gradient-delta weight. Rivero uses $\epsilon^{-d/2}$, so the corresponding squared limit acquires $(2\pi)^d$ under the present Fourier convention. This is a delta supported on the critical point, not a derivative of a delta. The argument fixes d; joint refinement requires control of the dimension-dependent weights and observables.

6 The two regulators and an explicit scaling map

Rivero's equation (5) contains

$$\exp\left(\frac{i\tau}{\epsilon} \sum_j L^\tau\left(q_j, \frac{q_{j+1} - q_j}{\tau}\right)\right),$$

and (7) takes the two regulators proportional. Under the literal physical identification $L^\tau = mv^2/2$ with fixed m and fixed physical coordinates, the exponent is iS_π/ϵ. Therefore $[\epsilon] =$ action, $[\tau] =$ time, and $\epsilon = b\tau$ assigns energy units to b. The positive Gaussian counterpart is precisely the finite-defect scaling in Proposition 2, with its path width tending to zero.

The source also permits changing L and the coordinate scale, and leaves its map (6) to be specified. Here is an exact free-model choice. Fix reference values $m_R > 0$, $\kappa_R > 0$ and set the bare kinetic coefficient to

$$m_B(\tau) = \frac{\epsilon(\tau)}{\kappa_R} m_R. \tag{9}$$

At every partition this yields

$$\frac{S_{\pi,B}}{\epsilon} = \frac{S_{\pi,R}}{\kappa_R}, \qquad \sqrt{\frac{m_B}{2\pi\epsilon\tau}} = \sqrt{\frac{m_R}{2\pi\kappa_R\tau}}.$$

Both the positive Gaussian family and its oscillatory counterpart are therefore fixed exactly, with the latter using i in the prefactor and the branch $1/\sqrt{i} = e^{-i\pi/4}$ for positive time. As $\epsilon \to 0$, this map sends $m_B \to 0$ while holding the reference mass m_R fixed. In the oscillatory case the bare phase corresponds to a fixed reference quantum coefficient; in the positive case the observable bridge variance is

$$\mathrm{Var}(q(s) \mid x, z) = \frac{\kappa_R s(T - s)}{m_R T}. \tag{10}$$

Choosing this variance at one interior time fixes the ratio κ_R/m_R; a separately fixed reference mass then fixes κ_R.

Wilson–Kogut [5, §12.2, pp. 166–168, Fig. 12.7] provides the design principle: compare cutoff theories at a common physical correlation length. Their construction specifies a renormalized mass and compares rescaled trajectories on the surface with dimensionless correlation length one. Under the topology discussed there, the renormalized mass remains a free parameter. In our model, (10) plays the role of the reference observable and (9) implements its preservation. This is an explicit toy scaling map, with the physical scale supplied by the renormalization condition.

7 The next physical selection test

The surviving information is now explicit: path locations, velocity oscillations, total excess action and normalized fluctuation width can have different limits. The next bounded task is to impose a physical dynamical condition on the oscillations and calculate which of these quantities survives. Uniform force control, a finite-speed relativistic action, and a prescribed fluctuation observable supply concrete alternative tests. For a universal positive action parameter, a selection argument must also relate different masses, durations and interacting systems, and explain the phase coefficient.

For the companion PDE question, this example motivates tracking the topology in which derivatives and quadratic quantities pass to a limit. The Yang–Mills target additionally concerns the physical Hamiltonian spectrum; the present defect is an action difference in a fixed-endpoint mechanics model.

References

[1] Victor Guillemin and Shlomo Sternberg. Semi-classical analysis. Author-hosted MIT notes, 2010. 13 January 2010; equation (14.7), pp. 463–465.

[2] Jim Pitman and Marc Yor. A guide to Brownian motion and related stochastic processes. arXiv:1802.09679v1, 2018.

[3] Filip Rindler. MA4G6 introduction to the modern calculus of variations. University of Warwick lecture notes, 2017. Spring 2017, version 2.0; Sections 2.1 and 3.3.

[4] Alejandro Rivero. A short derivation of Feynman formula. arXiv:quant-ph/9803035v1, 1998.

[5] Kenneth G. Wilson and John Kogut. The renormalization group and the ϵ expansion. *Physics Reports*, 12:75–199, 1974. DOI: 10.1016/0370-1573(74)90023-4.

A Positive Action Plateau from Classical Velocity Memory

navstokgap research working notes

7 September 2026

Abstract

A finite reversible classical velocity process determines a positive limiting action coefficient through its integrated velocity correlation. We define the coefficient operationally as mass times displacement variance divided by observation duration, derive its exact spectral formula and bounds, and compute a two-velocity example. Finite speed forces the short-duration coefficient to vanish while a positive long-duration plateau survives. This separates microscopic resolution from observation time and identifies the correlation law that a universal action-scale proposal must explain.

1 The observable and its classical premises

The candidate action field is an ensemble observable

$$\mathsf{h}_\Delta(t) = \frac{m}{\Delta}\operatorname{Var}[X(t+\Delta) - X(t)], \qquad m, \Delta > 0. \tag{1}$$

Here X is position on the line, t is physical time and Δ is an observation duration. Its units are mass times length squared per time, hence action. For Gaussian increments with variance $\kappa\Delta/m$, this normalization yields $\mathsf{h}_\Delta = \kappa$, matching the earlier free-kernel model. For stationary velocity statistics the observable is independent of t, and we write $\mathsf{h}(\Delta)$.

The present premises are classical stochastic ones: a finite set of velocities, ordinary probabilities, finite transition rates, a stationary ensemble and detailed balance. Their mathematical consequences give a positive transport scale. The project's stronger target adds a physical origin for these premises, universality across systems, and a derivation identifying the scale with $\hbar$. We reserve $h = 2\pi\hbar$ for Planck's constants and use h for the candidate observable. An independent dynamical field would require an additional state variable and its equation of motion.

2 Finite speed and the correlation formula

The observation duration determines which part of the velocity dynamics is resolved. In particular, bounded velocity gives a sharp elementary constraint.

Proposition 1 (Microscopic limit)**.** *If X has absolutely continuous paths with $|\dot{X}| \le u < c$ almost everywhere, almost surely, then*

$$0 \le \mathsf{h}_\Delta(t) \le m u^2 \Delta \longrightarrow 0 \quad (\Delta \downarrow 0).$$

Proof. The displacement has absolute value at most $u\Delta$. Its variance is at most its second moment, which is at most $u^2\Delta^2$. Divide by Δ/m. □

For a real centered stationary velocity V with finite second moment, set $X(t) = \int_0^t V(s)\,ds$ and $C(s) = \mathbb{E}[V(s)V(0)]$. Fubini's theorem on each finite square is justified by Cauchy–Schwarz and stationarity. Symmetry of the covariance gives

$$\mathsf{h}(\Delta) = 2m \int_0^\Delta (1 - s/\Delta) C(s)\,ds. \tag{2}$$

If $C \in L^1(0,\infty)$, dominated convergence yields

$$H_* := \lim_{\Delta\to\infty} \mathsf{h}(\Delta) = 2m \int_0^\infty C(s)\,ds = 2mD. \tag{3}$$

This is the Green–Kubo diffusion coefficient D, with an action normalization [4, (1.3), Proposition 2.1]. The finite-state assumptions below make its strict positivity and convergence explicit.

3 A positive plateau from finite reversible dynamics

Let J_t be a stationary irreducible continuous-time Markov chain on $\{1,\dots,k\}$, $k \geq 2$, with invariant probabilities $\pi_i > 0$ and generator Q. Thus $Q_{ij} \geq 0$ for $i \neq j$, $Q\mathbf{1} = 0$ and $\pi_i Q_{ij} = \pi_j Q_{ji}$. All rates are finite. Assign velocities v_i with

$$\sum_i \pi_i v_i = 0, \qquad \sigma_v^2 := \sum_i \pi_i v_i^2 > 0, \qquad |v_i| \leq u < c.$$

Set $V(t) = v_{J_t}$ and integrate to obtain X. The Markov state is (X, J); the velocity label retains the information required for future motion. Paths are piecewise linear, with finitely many velocity jumps on compact intervals almost surely. These jumps represent impulsive interactions rather than a smooth prescribed force law.

Use $\langle f, g\rangle_\pi = \sum_i \pi_i f_i g_i$. On the mean-zero subspace, $-Q$ is positive definite: detailed balance gives

$$\langle f, -Qf\rangle_\pi = \tfrac{1}{2} \sum_{i,j} \pi_i Q_{ij} (f_j - f_i)^2,$$

and irreducibility makes its nullspace consist of constants. Let its positive eigenvalues be γ_j, with an orthonormal eigenbasis e_j, and write $v = \sum_{j=1}^{k-1} a_j e_j$.

Proposition 2 (Classical action plateau)**.** *Under these premises,*

$$\mathsf{h}(\Delta) = 2m \sum_{j=1}^{k-1} \frac{a_j^2}{\gamma_j} \left[1 - \frac{1 - e^{-\gamma_j \Delta}}{\gamma_j \Delta}\right], \tag{4}$$

$$H_* = 2m\langle v, (-Q)^{-1} v\rangle_\pi = 2m \sum_{j=1}^{k-1} \frac{a_j^2}{\gamma_j} > 0. \tag{5}$$

The inverse acts on mean-zero functions. Writing $\gamma_{\min} = \min_j \gamma_j$ and $\gamma_{\max} = \max_j \gamma_j$ gives

$$\frac{2m\sigma_v^2}{\gamma_{\max}} \leq H_* \leq \frac{2m\sigma_v^2}{\gamma_{\min}}, \qquad 0 \leq 1 - \frac{\mathsf{h}(\Delta)}{H_*} \leq \frac{1}{\gamma_{\min}\Delta}. \tag{6}$$

Moreover $\mathsf{h}(\Delta)$ increases strictly from zero to H_*; in particular $\mathsf{h}(\Delta) \geq H_*/2$ for $\Delta \geq 2/\gamma_{\min}$.

Proof. The transition semigroup gives $C(s) = \langle v, e^{sQ} v\rangle_\pi = \sum_j a_j^2 e^{-\gamma_j s}$. Inserting this into (2) and integrating each exponential proves (4). At least one a_j is nonzero, so the limit is strictly positive. The identity $\sum_j a_j^2 = \sigma_v^2$ gives the two spectral bounds. Each relative correction is bounded above by $1/(\gamma_{\min}\Delta)$, which gives the convergence estimate. Finally the scalar function $f(y) = 1 - (1 - e^{-y})/y$ satisfies $f'(y) = [1 - (1+y)e^{-y}]/y^2 > 0$ for $y > 0$, since $e^y > 1 + y$. Its endpoint limits are zero and one. Positive weighted summation proves strict monotonicity and the stated duration-dependent bound. $\square$

This is a finite-state spectral specialization of the generator Poisson representation in [4, §2]. The bounds concern the observable (1) and fixed model parameters. Within the same axiom class, replacing Q by rQ for $r > 0$ changes H_* to H_*/r. Thus a uniform lower bound over models requires control of the transition rates as well as nonzero velocity variance. The spectral gap of the velocity generator has inverse-time units; H_* has action units.

4 Two velocities: the full time dependence

Choose velocities $\pm u$, $0 < u < c$, each with stationary probability 1/2, and a reversal rate $\lambda > 0$:

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \lambda & -\lambda \end{pmatrix}, \qquad V(t) = u\sigma_0(-1)^{N_t}.$$

Here N_t is a Poisson counting process of rate λ, independent of the uniform sign σ_0. Its generating function gives $\mathbb{E}[(-1)^{N_s}] = e^{-2\lambda s}$, hence

$$C(s) = u^2 e^{-2\lambda s}, \qquad \mathsf{h}(\Delta) = \frac{mu^2}{\lambda}\left[1 - \frac{1 - e^{-2\lambda\Delta}}{2\lambda\Delta}\right]. \tag{7}$$

Therefore

$$\mathsf{h}(\Delta) \sim mu^2\Delta \quad (\Delta \downarrow 0), \qquad H_* = \frac{mu^2}{\lambda} > 0 \quad (\Delta \to \infty).$$

The correlation time is $1/(2\lambda)$, so the limiting action is twice mass times velocity variance times correlation time. A particle starting at $X(0) = 0$ has support inside $[-u\Delta, u\Delta]$ at time Δ; the no-reversal events give atoms at both endpoints. The model uses probability measures, including these atoms, rather than assuming a Gaussian density at every duration.

The velocity-resolved position measures $p_\pm(A, t) = \Pr\{X(t) \in A, V(t) = \pm u\}$ satisfy, distributionally,

$$\partial_t p_+ = -u\partial_x p_+ - \lambda p_+ + \lambda p_-, \qquad \partial_t p_- = u\partial_x p_- + \lambda p_+ - \lambda p_-.$$

Adding and subtracting these equations and eliminating their difference gives the telegraph equation for $p = p_+ + p_-$,

$$\partial_t^2 p + 2\lambda\partial_t p = u^2\partial_x^2 p.$$

This is the persistent-flight construction associated with the classical telegraph process [1, pp. 1–2, (1.1)–(1.3)].

5 Which limit keeps the action?

Microscopic resolution and diffusion scaling give different values of the same observable. Fix $a > 0$ with diffusion units and take the formal family $u_\lambda^2 = a\lambda$. For every fixed $\Delta > 0$,

$$\mathsf{h}_\lambda(\Delta) = ma\left[1 - \frac{1 - e^{-2\lambda\Delta}}{2\lambda\Delta}\right] \longrightarrow ma.$$

Consequently the iterated limits are

$$\lim_{\Delta\downarrow 0}\lim_{\lambda\to\infty} \mathsf{h}_\lambda(\Delta) = ma, \qquad \lim_{\lambda\to\infty}\lim_{\Delta\downarrow 0} \mathsf{h}_\lambda(\Delta) = 0. \tag{8}$$

This is an exact statement about second moments. The formal scaling sends $u_\lambda \to \infty$ and therefore leaves any fixed physical speed ceiling. At fixed $u < c$ and fixed λ, the positive plateau instead occurs for observation durations long compared with the correlation time.

6 Axiom audit and the universality question

Each premise has a visible mathematical role:

Classical premise	Consequence in the proof
Stationary centered ensemble	Correlation depends on elapsed time; mean drift is removed
Finite irreducible transition dynamics	Inverse generator exists on mean-zero functions
Detailed balance	Real orthogonal velocity modes with positive decay rates
Nonzero velocity variance	At least one mode contributes to the positive plateau
Finite speed	Short-duration action coefficient tends to zero
Mass and transition rates	Set the numerical value and units of the plateau

These premises establish positivity for each model in the class. Independent composition permits particles with different transition rates, so universality adds a further dynamical requirement. For the two-velocity family, a common limiting value H_* across masses requires

$$\lambda_m = \frac{m u_m^2}{H_*}. \tag{9}$$

Equation (9) states the rate law to explain. It is not supplied by independent composition. Likewise, the present effective model specifies impulsive reversals externally; a closed mechanical realization must account for their momentum exchange and correlation law.

Two reconstruction sources sharpen this requirement. Nelson's stated Brownian hypothesis supplies $D = \hbar/(2m)$ at the outset [3, abstract]. Hall and Reginatto instead add momentum fluctuations whose strength scales inversely with position uncertainty; their ensemble coefficient C is identified with $(\hbar/2)^2$ [2, pp. 5, 8]. These are concrete comparison premises for the origin of a shared scale.

The next research step is to test whether a mechanical realization can enforce the mass-weighted correlation integral in (3) as a shared invariant. That would advance the scale-selection target beyond a family of independent transport coefficients. A separate phase/composition argument must then identify the common constant with $\hbar$.

References

[1] Fabrizio Cinque. A note on the conditional probabilities of the telegraph process. arXiv:2202.01904v1, 2022. Version 1; pp. 1–2.

[2] Michael J. W. Hall and Marcel Reginatto. Schrödinger equation from an exact uncertainty principle. arXiv:quant-ph/0102069v3, 2002. Version 3; selected pp. 1–5 and 8.

[3] Edward Nelson. Derivation of the Schrödinger equation from Newtonian mechanics. *Physical Review*, 150:1079, 1966. Official abstract consulted.

[4] G. A. Pavliotis. Asymptotic analysis of the Green–Kubo formula. arXiv:1002.4103v1, 2010. Preprint version 1; equation (1.3) and Section 2.

Physical-Time Action Relaxation in an Elastic Collision Bath

navstokgap research working notes

7 September 2026

Abstract

An elastic repeated-collision model produces an action-valued covariance observable that relaxes exponentially in physical time to a positive constant. We derive its rate and plateau directly from the masses, bath velocity variance and collision frequency. Bounded bath velocities keep every collision below a fixed speed ceiling. The exact parameter dependence isolates the additional law needed for a universal action scale: energy and momentum conservation determine the collision map, while the reservoir statistics set the plateau.

1 A momentum receiver and a classical stochastic clock

A tracer of mass $m \geq M > 0$ meets fresh bath particles of mass M on the line. Collisions occur at an independent Poisson rate $\nu > 0$. Incoming bath velocities W_n are independent, identically distributed, centered and obey $|W_n| \leq w < c/3$, with variance $s^2 > 0$. Initial tracer velocity is independent of the bath and clock, centered, and bounded by w. Between collisions it is constant; $X(t) = X(0) + \int_0^t V(r)\,dr$.

The nontrivial one-dimensional elastic collision map is

$$V' = aV + bW, \quad W' = \frac{2m}{m+M}V - aW, \qquad a = \frac{m-M}{m+M}, \quad b = \frac{2M}{m+M} = 1 - a. \tag{1}$$

Direct substitution gives $mV' + MW' = mV + MW$ and $m(V')^2 + M(W')^2 = mV^2 + MW^2$. Since $0 \leq a < 1$, the tracer update is a convex combination and preserves $|V| \leq w$. The departing particle obeys $|W'| \leq (3m - M)w/(m + M) < 3w < c$.

The model specifies a refreshed reservoir and a velocity-independent clock. These are Barkai's collision-model premises [2, pp. 3–5, (2)–(12)], specialized here to a bounded bath and $m \geq M$. Each collision accounts for momentum and energy transfer; the complete stream supplies fresh particles and removes departing ones. A spatial gas realization would additionally derive encounter rates and incoming velocity statistics. The speed ceiling here is a kinematic restriction on Newtonian collisions in the reservoir frame, rather than a Lorentz-covariant collision law.

2 Stationary covariance and the action plateau

Define the two inverse-time rates

$$\gamma = \nu(1-a) = \frac{2\nu M}{m+M}, \qquad \beta = \nu(1-a^2) = \frac{4\nu mM}{(m+M)^2}. \tag{2}$$

The velocity generator acts on bounded functions as $Lf(v) = \nu\{\mathbb{E}[f(av + bW)] - f(v)\}$.

Proposition 1 (Collision plateau)**.** *The stationary velocity law is unique on* $[-w, w]$ *and is represented by the convergent series* $V_* \overset{d}{=} b\sum_{j=0}^{\infty} a^j W_j$*. Its variance and autocovariance are*

$$S_* := \operatorname{Var} V_* = \frac{b^2 s^2}{1-a^2} = \frac{M}{m} s^2, \qquad C(r) = S_* e^{-\gamma r}. \tag{3}$$

Thus the stationary displacement observable $\mathsf{h}(\Delta) = m \operatorname{Var}[X(t+\Delta) - X(t)]/\Delta$ *satisfies*

$$\mathsf{h}(\Delta) = H_* \left[1 - \frac{1 - e^{-\gamma\Delta}}{\gamma\Delta}\right], \qquad H_* := \frac{2mS_*}{\gamma} = \frac{(m+M)s^2}{\nu} > 0. \tag{4}$$

Proof. The series converges absolutely and stays in $[-w, w]$, since $b\sum a^j = 1$ (for $a = 0$ it is simply W_0). Separating its first term proves invariance under one collision. Coupling two initial velocities to the same clock and bath multiplies their difference by a^{N_t}. Its mean contraction factor is $\mathbb{E}a^{N_t} = e^{-\nu(1-a)t}$, also for $a = 0$ with $0^0 = 1$. Hence invariant probability laws on the bounded interval coincide. Independence and centering give the variance in (3). Conditioning on the collision count yields $\mathbb{E}[V(t+r) \mid V(t)] = e^{-\gamma r}V(t)$; stationary covariance follows. Integrating the covariance over the displacement square gives $\mathsf{h}(\Delta) = 2m\int_0^\Delta (1 - r/\Delta)C(r)\,dr$, which evaluates to (4). □

This proof uses a contracting affine collision update; it requires neither a finite velocity state space nor detailed balance of the refreshed bath. The energy-valued reservoir parameter $\Theta := Ms^2$ gives $mS_* = \Theta$ and $H_* = \Theta(m+M)/(\nu M)$. For a general bounded bath, Θ denotes twice the mean incoming kinetic energy, without assuming a thermodynamic equilibrium distribution. The stationary covariance and Green–Kubo integral also follow from Barbier and Trizac's field-free Maxwell-kernel specialization [1, p. 19, (89)–(98)]. Their kernel exponent called ν is zero there; our ν is a positive collision frequency.

3 An observable relaxing in physical time

The integrated future-lag covariance defines a second operational quantity:

$$\mathsf{a}(t) := 2m \int_0^\infty \operatorname{Cov}[V(t+r), V(t)]\,dr. \tag{5}$$

It has action units and uses physical preparation time t, with the lag integrated out. Its definition differs from the finite-duration displacement observable; both share the same stationary long-duration plateau.

Proposition 2 (Physical-time relaxation)**.** *For the centered initial ensemble specified above, with* $S_0 = \operatorname{Var} V(0)$*,*

$$S(t) = S_* + (S_0 - S_*)e^{-\beta t}, \tag{6}$$

$$\mathsf{a}(t) = \frac{2mS(t)}{\gamma} = H_* + (\mathsf{a}(0) - H_*)e^{-\beta t}, \qquad \dot{\mathsf{a}} = \beta(H_* - \mathsf{a}). \tag{7}$$

In particular a tracer prepared at rest has $\mathsf{a}(t) = H_*(1 - e^{-\beta t}) > 0$ *for every* $t > 0$ *and* $\mathsf{a}(t) \to H_* > 0$ *as* $t \to \infty$*.*

Proof. The generator gives $Lv = -\gamma v$ and $L(v^2) = -\beta v^2 + \nu b^2 s^2$. Centering is preserved, so $\dot{S} = -\beta S + \nu b^2 s^2 = \beta(S_* - S)$. The conditional mean from Proposition 1 gives, even away from stationarity, $\operatorname{Cov}[V(t+r), V(t)] = e^{-\gamma r}S(t)$. Integration proves (7). □

The original displacement definition retains a distinct duration parameter:

$$\mathsf{h}_\Delta(t) = \frac{2m}{\gamma\Delta}\int_0^\Delta S(t+r)\big[1-e^{-\gamma(\Delta-r)}\big]\,dr. \tag{8}$$

Indeed, for $r \le s$ the covariance inside its double integral is $e^{-\gamma(s-r)}S(t+r)$; integrating over s proves the formula. At fixed Δ, it approaches the stationary value in (4) exponentially at rate β. At every fixed t it tends to zero as $\Delta \downarrow 0$, with $\mathsf{h}_\Delta(t) \sim mS(t)\Delta$ when $S(t) > 0$. Its long-duration limit is H_*, since the transient contribution to (8) is bounded in magnitude by $2m|S_0 - S_*|e^{-\beta t}/(\gamma\beta\Delta)$.

4 What would select a common constant?

For fixed bath mass, variance and clock, the plateau grows as $m + M$. Achieving the same H_* for different tracer masses requires

$$\nu_m = \frac{(m+M)s_m^2}{H_*}. \tag{9}$$

This equation is a diagnostic of a proposed encounter law. Binary energy and momentum conservation alone place no restriction of this form on the external clock or reservoir preparation.

More explicitly, take a fixed admissible bath and a tracer initially at rest. For any $0 < \epsilon \le 1$, replace every incoming velocity by $W_n^{(\epsilon)} = \epsilon W_n$, keeping masses and clock unchanged. Every elastic collision, centering assumption and speed bound remains valid. Both H_* and $\mathsf{a}(t)$ are multiplied by ϵ^2. Consequently this entire class admits arbitrarily small positive plateaus even though every nondegenerate member relaxes to its own positive constant. This is a countermodel test for these specified stochastic collision premises.

The next model should replace the supplied clock by a spatial encounter law and test how it scales with reservoir velocity and density. The physical-time relaxation mechanism is already explicit: persistent injection of velocity variance balances collisional contraction. A universal quantum action scale requires an additional physical relation that fixes this balance across systems.

Ben-Naim and Krapivsky provide a useful clock comparison: exponential decay in collision number becomes algebraic decay in physical time in their cooling model [3, p. 6, (14)–(17)].

5 Return to the partition question

The project's central question concerns a sequence of time partitions $\pi_N = \{0 = t_0^{(N)} < \cdots < t_N^{(N)} = T\}$ with mesh tending to zero. The present relaxation law describes preparation time under a supplied bath. It therefore serves as a comparison mechanism for the partition problem: one must derive, from admissible cut-point refinements, both the quantity that survives and the condition selecting its positive value. The collision calculation isolates how a positive constant can instead enter through reservoir energy and a clock. The next partition test tracks these inputs explicitly when a refinement law is proposed.

References

[1] Matthieu Barbier and Emmanuel Trizac. Field induced stationary state for an accelerated tracer in a bath. arXiv:1203.6759v3, 2012. Selected pp. 1, 19; (89)–(98).

[2] Eli Barkai. Stable equilibrium based on Lévy statistics: A linear Boltzmann equation approach. arXiv:cond-mat/0303255v1, 2003. Selected pp. 1, 3–5.

[3] E. Ben-Naim and P. L. Krapivsky. The inelastic Maxwell model. arXiv:cond-mat/0301238v1, 2003. Selected pp. 1, 6; (14)–(17).

A surviving action defect and consistency under inserting cuts

A03, reviewed with B12, 2026-09-07. Two exact tests separate refinement of one experiment from a changing family of fluctuation preparations. Constant-force chord errors vanish on every mesh tending to zero. Within the scalar Gaussian bridge family, exact consistency under retaining old nodes fixes the fluctuation parameter. These statements locate the next selection obligation at the refinement law.

1. Arbitrary partitions of the constant-force trajectory

Fix $m, F, T > 0$, $x(t) = v_0 t$ with $v_0 > 0$, $y(t) = Ft^2/(2m)$, and the Lagrangian $L = m(\dot{x}^2 + \dot{y}^2)/2 + Fy$. Let $\pi = (0 = t_0 < \cdots < t_N = T)$, $\tau_j = t_j - t_{j-1}$, and let q_π linearly interpolate the exact trajectory at these times. Both global endpoints and all sampled positions are fixed in this comparison.

On interval j, with $s = t - t_{j-1}$, the interpolation error is $\eta_j(s) = Fs(\tau_j - s)/(2m)$. The classical first variation vanishes on that interval. Adding the exact local quadratic action differences gives

$$D_\pi := S[q_\pi] - S[q_{\mathrm{cl}}] = \frac{F^2}{24m} \sum_j \tau_j^3 \le \frac{F^2 T}{24m} |\pi|^2 \longrightarrow 0.$$

The unsigned chord–arc lens areas similarly add to $A_\pi = v_0 F \sum_j \tau_j^3/(12m)$, so $D_\pi = FA_\pi/(2v_0)$. The existing constant-force calculation applies after subtracting the local linear tangent; its slope cancels from the lens difference.

For fixed N, convexity yields $\sum_j \tau_j^3 \ge T^3/N^2$, attained by equal steps. Splitting one interval into positive lengths a, b reduces the action defect by

$$\frac{F^2}{24m}[(a+b)^3 - a^3 - b^3] = \frac{F^2}{8m} ab(a+b) > 0.$$

Thus nonuniform cuts cannot defeat convergence when the maximum step tends to zero. This calculation concerns sampled chords; Newton's impulsive polygon is a separate approximation scheme requiring its own error estimate.

2. Exact consistency of Gaussian node laws

Fix the free-line experiment on $[0, T]$, mass $m > 0$ and fixed endpoints. For each partition with at least one internal node, let the centered node vector be Gaussian with covariance

$$\mathrm{Cov}_\pi(\eta_i, \eta_j) = \frac{\kappa_\pi}{m} G(t_i, t_j), \qquad G(s,t) = \min(s,t) - st/T, \quad \kappa_\pi \ge 0.$$

The zero value denotes a point mass at the classical straight path. Exact refinement consistency means that when $\pi' \supset \pi$, deleting the new coordinates from the finer law gives precisely the coarser law, with the same physical times, positions and mass.

Proposition. Within this family, consistency forces $\kappa_{\pi'} = \kappa_\pi$ for every refinement retaining an interior node. Conversely a fixed κ gives consistent Gaussian node laws.

Proof. At any retained interior time s, equality of the one-node marginals implies $\kappa_{\pi'} s(T - s)/(mT) = \kappa_\pi s(T-s)/(mT)$. The factor multiplying κ is positive. Conversely the restriction of a centered Gaussian vector has the corresponding covariance submatrix, which here is exactly the coarse covariance. On the directed family of all partitions, any two nontrivial partitions have a common refinement, so their parameters agree. The endpoint-only partition has no observable variance and places no extra constraint. This completes the proof.

For nested π_N retaining an interior point s, M05's finite-defect scaling $(N-1)\kappa_N \to \ell > 0$ makes that retained variance approach zero. It therefore describes changing preparations, rather than exact marginals of one positive-width bridge experiment. It remains a valid triangular-array limit, with the action defect proved in C014.

3. What survives at fixed preparation?

Let $d = N - 1$. Under the fixed-$\kappa > 0$ bridge, the polygon's free action excess satisfies $2D_\pi/\kappa \sim \chi_d^2$, independent of the actual positive step lengths (C014's finite law). Consequently

$$\mathbb{E}D_\pi = d\kappa/2, \qquad \operatorname{Var} D_\pi = d\kappa^2/2,$$

and $D_\pi \to \infty$ in probability as $d \to \infty$. Indeed, Chebyshev gives $\Pr\{D_\pi < d\kappa/4\} \leq 8/d$. The normalized estimator

$$\widehat{\kappa}_\pi = \frac{2D_\pi}{d}$$

has mean κ and variance $2\kappa^2/d$, hence converges to κ in mean square, even on nonuniform partitions. It estimates the supplied bridge parameter. It is not an additive accumulated action; division by the number of internal nodes is essential. For $\kappa = 0$ every defect vanishes.

Thus exact consistency permits a fixed positive action parameter and a consistent estimator of it, while the unnormalized kinetic action diverges. It also permits the deterministic zero family. Selection of a positive scale still requires a physical premise excluding that family and fixing the scale.

4. One inserted cut and the fluctuation it introduces

On a coarse interval of length $a + b$, fix endpoint positions x, z and insert y after duration a, with $a, b > 0$. Set $\bar{y} = (bx + az)/(a + b)$ and $\zeta = y - \bar{y}$. Completing the square gives

$$\frac{m}{2}\left[\frac{(y-x)^2}{a} + \frac{(z-y)^2}{b} - \frac{(z-x)^2}{a+b}\right] = \frac{m(a+b)}{2ab}\zeta^2.$$

At fixed bridge parameter κ, conditional Gaussian variance is $\operatorname{Var}(\zeta \mid x, z) = \kappa ab/[m(a+b)]$. The expected extra action from inserting one node is therefore $\kappa/2$, independent of the interval lengths. For deterministic interpolation $\zeta = 0$ it is zero. The kinetic splitting identity is nonnegative, whereas section 1's total constant-force chord error decreases under refinement: the latter includes the potential term and moves the inserted node onto the accelerated classical curve.

This is a local, testable formulation of the missing premise: does inserting a physical cut introduce a nonzero conditional fluctuation, and what law sets its variance? Supplying the Gaussian rule answers the first question by assumption. A classical derivation would have to produce that rule or a different consistent fluctuation mechanism from independent physics.

5. Next test and prior-art gate

The B12 audit classifies C027–C029 as elementary consequences. Pitman and Yor, *A guide to Brownian motion and related stochastic processes* (2018), arXiv:1802.09679v1, printed pp. 6, 10, 12–13, supply the Gaussian bridge and restriction framework. The source companion records coverage. The constant-force identity extends the existing calculation. The action law and moments in section 3 are proved in the regulator-limits paper. The combined selection test is the research use of these established ingredients.

Next distinguish coordinate sampling from a physically executed intervention at a cut. A candidate intervention law must state its effect on energy, momentum, conditional variance and coarse observables. Test consistency first, then positivity, scale universality and compatibility with finite speed. The Gaussian conditional law is a reference model for that audit, not yet a model with a hard physical speed ceiling.

A physical cut: elastic reversal, finite speed and memory

A05, reviewed with B13, 2026-09-07. An equal-mass elastic collision realizes a random midpoint with exact energy and momentum conservation. Its action cost tends to zero with the interval duration at fixed speed. A sharp midpoint bound extends this conclusion to every bounded-speed bridge law. Finally, a position-only convolution law with ballistic support is necessarily a deterministic drift. Together these tests identify velocity memory as a concrete ingredient for the next refinement model.

1. One collision with a momentum receiver

Fix $m > 0$, $0 < u < c$ and duration $\Delta > 0$. Choose a common random sign $\sigma = \pm 1$ with equal probabilities. At time zero, a labelled tracer is at 0 with velocity σu, and a second particle of the same mass is at $\sigma u \Delta$ with velocity $-\sigma u$. Both move freely on the line until their instantaneous elastic collision at time $\Delta/2$. The equal-mass collision exchanges their velocities. Thus the tracer follows

$$X(t) = \begin{cases} \sigma u t, & 0 \le t \le \Delta/2, \\ \sigma u(\Delta - t), & \Delta/2 \le t \le \Delta. \end{cases}$$

Its midpoint $Y = \sigma u \Delta/2$ is random, while $X(0) = X(\Delta) = 0$. The other particle returns to its own initial position. Throughout the motion, total momentum is zero and total kinetic energy is mu^2. Each labelled velocity changes sign at the collision, with opposite impulses of magnitude $2mu$. Both particles stay below c in the chosen frame.

The tracer's free kinetic action relative to the stationary path with the same position endpoints is

$$D = \frac{m}{2} \int_0^\Delta \dot{X}^2 dt = \frac{mu^2 \Delta}{2}. \qquad \operatorname{Var} Y = \frac{u^2 \Delta^2}{4}.$$

Define the midpoint action parameter in A03's normalization by $\kappa_{\text{mid}} = 4m \operatorname{Var} Y/\Delta$. This gives $\kappa_{\text{mid}} = mu^2 \Delta$ and $\mathbb{E}D = \kappa_{\text{mid}}/2$. At fixed energy and speed, both tend to zero as $\Delta \downarrow 0$. Keeping a supplied $\kappa_{\text{mid}} > 0$ instead requires $u^2 = \kappa_{\text{mid}}/(m\Delta)$ and pair energy $\kappa_{\text{mid}}/\Delta$.

This is an exactly specified hard-collision model with random initial data. The timing follows from the prepared initial separation $u\Delta$, rather than a new stochastic clock. Changing Δ changes that separation. The stationary comparison shares positions and times, not initial velocities or apparatus state. Consequently the construction preserves the coarse position endpoints, but does not implement an invisible intervention on a fixed full phase-space preparation. Smooth collision potentials and a Lorentz-covariant dynamics are separate models.

2. Sharp midpoint support and variance bound

Let any absolutely continuous path on $[0, \Delta]$ have endpoints x, z and $|\dot{X}| \le u < c$ almost everywhere. Write $v = (z - x)/\Delta$, with $|v| \le u$, and $\zeta = X(\Delta/2) - (x + z)/2$. Each half-interval gives

$$X(\Delta/2) \in [x - u\Delta/2, x + u\Delta/2] \cap [z - u\Delta/2, z + u\Delta/2].$$

This intersection is centered at $(x + z)/2$ with half-width $r = \Delta(u - |v|)/2$. Therefore any probability law on these admissible paths satisfies

$$\operatorname{Var} Y \le r^2, \qquad 0 \le \kappa_{\text{mid}} := \frac{4m}{\Delta} \operatorname{Var} Y \le m\Delta(u - |v|)^2.$$

Proof of the variance bound: $\operatorname{Var} Y \leq \mathbb{E}(Y-(x+z)/2)^2 \leq r^2$. Equal probabilities at the two extremal midpoints saturate it; the corresponding two-segment paths obey the speed bound. For $x = z$ the elastic example realizes this sharp case.

The two-segment interpolant through Y has excess kinetic action $D_{\text{poly}} = 2m\zeta^2/\Delta$. When the midpoint law is centered on $(x+z)/2$, $\mathbb{E}D_{\text{poly}} = \kappa_{\text{mid}}/2$. For a biased midpoint there is the additional term $2m(\mathbb{E}\zeta)^2/\Delta$. This distinction avoids identifying a variance with all action costs.

A Gaussian midpoint with variance $\kappa\Delta/(4m)$ has unbounded support for every $\kappa > 0$ and so fails the exact speed constraint at every duration. Even a bounded replacement matching only that variance requires $\kappa \leq m\Delta(u-|v|)^2$. The latter inequality gives a necessary resolution restriction, not a derivation of κ.

At $|v| = u$ the allowable midpoint is unique. Indeed a path achieving the maximum displacement $u\Delta$ must have velocity u almost everywhere. Thus after choosing an extremal midpoint in the return-path example, each half is already a saturated straight segment: finer admissible sampling inside that half is deterministic. Reapplying a nonzero midpoint fluctuation rule independently at every cut would change the coarse law or violate speed.

3. Finite speed and a position-only convolution law

Proposition. Suppose probability measures μ_t on the line satisfy $\mu_0 = \delta_0$, $\mu_{s+t} = \mu_s * \mu_t$, and $\operatorname{supp} \mu_t \subset [-ut, ut]$ for every $t \geq 0$, with a fixed finite u. Then $\mu_t = \delta_{bt}$ for some $|b| \leq u$.

Proof. For fixed $T > 0$ and every positive integer n, convolution and bounded support give

$$\operatorname{Var} \mu_T = n \operatorname{Var} \mu_{T/n} \leq nu^2(T/n)^2 = u^2T^2/n.$$

Let $n \to \infty$ to obtain zero variance at every T. Write $\mu_t = \delta_{a(t)}$. The convolution law makes a additive, while $|a(t)| \leq ut$ implies continuity at zero. Hence $a(t) = bt$, $|b| \leq u$. This proof even derives the needed continuity from the support bound.

The proposition concerns stationary, spatially homogeneous, independent increment composition on position alone. It leaves room for stochastic bounded-speed trajectories with velocity memory. For example A01's telegraph process is Markov in (X, V); deleting V removes the information needed for such position-only convolution. General state-dependent Markov motion is also outside the stated convolution hypothesis.

4. Consequence for the cut-point programme

Finite propagation does not supply a positive microscopic midpoint action coefficient in these models. It identifies a structural requirement: a stochastic refinement law with strict speed must retain correlations or additional state information, rather than resetting independent position increments. The elastic construction makes this visible at a single cut.

The next bounded task is to derive a finite-speed bridge on (X, V), including its conditional midpoint law and endpoint atoms, and compare coarse and fine restrictions of the same experiment. Then separate a finite observation-window action scale from a microscopic refinement remainder. The universal positive scale remains a selection question about the physical premises.

Proof and literature record

The B13 audit reviews C030–C032 as derived consequences. Barkai, *Stable Equilibrium Based on Lévy Statistics* (2003), arXiv:cond-mat/0303255v1, pp. 3–4, (2)–(3), supplies the elastic map. The equal-mass map is well-defined even though the divided-density representation (5) excludes that mass ratio. Cinque, *A Note on the Conditional Probabilities of the Telegraph Process* (2022), arXiv:2202.01904v1, pp. 1–2, supplies the velocity-memory and endpoint-atom comparison. The midpoint and convolution proofs above follow directly from their stated hypotheses;

neither selected source states those combined results. The source companion records the bounded coverage. No priority claim follows from this audit.

A finite-speed return bridge and its polygonal action

A velocity-resolved telegraph bridge gives one consistent random trajectory at every cut resolution. Its midpoint has a discrete probability mass, while the action error of sampled polygons vanishes at a controlled rate. The construction isolates velocity memory from the introduction of fresh noise at each cut.

1. Model and endpoint convention

Fix mass $m > 0$, speed $0 < u < c$, reversal rate $\lambda > 0$ and duration $T > 0$. In an inertial frame set $X_0 = 0$, $V_0 = u$ and

$$V_t = u(-1)^{N_t}, \qquad X_t = \int_0^t V_s\,ds,$$

where N is a homogeneous Poisson process of rate λ. Velocity is right-continuous. The rate is a supplied classical stochastic clock. This reduced model describes impulsive reversals; momentum receivers require the extra interaction bookkeeping of the physical-cut model (C030).

We construct the bridge to $(X_T, V_T) = (0, -u)$ by disintegrating switch-time densities at the interior position $X_T = 0$. The explicit simplex construction below fixes the version of this zero-probability conditioning. Before conditioning, the zero-switch event has probability $e^{-\lambda T}$ and lies at $(uT, +u)$; the chosen terminal velocity selects odd positive switch counts.

The free kinetic functional is $S[X] = (m/2)\int_0^T V_t^2 dt$, in action units. We compare it with the sampled polygon and with the zero-position chord. The chord matches position endpoints; its velocity endpoints differ from the bridge. All comparisons below name this positional endpoint convention.

2. Exact conditional path law

Write $N_T = 2K + 1$ and $z = \lambda T$. The bridge count law is

$$\boxed{\Pr(K = k \mid \text{bridge}) = w_k = \frac{(z/2)^{2k}}{(k!)^2 I_0(z)}, \qquad k = 0, 1, \ldots,}$$

where $I_0(z) = \sum_{k\geq 0}(z/2)^{2k}/(k!)^2$. Given $K = k$, independently choose uniform simplex vectors $(P_0, \ldots, P_k)$ and $(M_0, \ldots, M_k)$, each summing to $T/2$. Follow velocities $+u, -u, +u, -u, \ldots$ for durations $P_0, M_0, P_1, M_1, \ldots, P_k, M_k$. For $k = 0$ each simplex is a singleton. This specifies the entire conditioned path, including all later observations.

Derivation. Given $N_T = 2k + 1$, the ordered switch times are uniform in $0 < s_1 < \cdots < s_{2k+1} < T$. Their $2k + 2$ spacings are uniform on the simplex of total length T. The total positive duration A has density

$$f_k(a) = \frac{(2k+1)!}{(k!)^2 T^{2k+1}} a^k (T - a)^k, \qquad 0 < a < T.$$

Since $X_T = u(2A - T)$, the position Jacobian is $da/dx = 1/(2u)$. Multiplication by the Poisson count weight gives the joint density

$$\left.\frac{\Pr(X_T \in dx, N_T = 2k+1)}{dx}\right|_{x=0} = \frac{e^{-\lambda T}\lambda^{2k+1}}{2u(k!)^2}(T/2)^{2k}.$$

Its sum is $e^{-\lambda T}\lambda I_0(z)/(2u) > 0$. Division gives w_k. Conditioning the uniform spacings on $A = T/2$ leaves the two independent simplexes stated above. The occupation-time backbone is the equal-rate specialization of Cinque's Theorem 2.1, (2.6), printed p. 4; the source also gives the alternating switch representation (2.4), p. 3.

3. Midpoint law, including the atom

The exact midpoint law is an explicit pushforward of the preceding mixture. Let $d = (P_0, M_0, \ldots, P_k, M_k)$, $s_0 = 0$ and $s_j = \sum_{i<j} d_i$. Then

$$Y = X_{T/2} = u \sum_{j=0}^{2k+1} (-1)^j \min\{d_j, \max(0, T/2 - s_j)\}.$$

For any bounded measurable g, its expectation is $\sum_k w_k \mathbb{E}_k[g(Y)]$, where $\mathbb{E}_k$ is integration over the two normalized simplexes. This formula sums over midpoint velocities. The joint law retains $V_{T/2} = u(-1)^j$ on $s_j \le T/2 < s_{j+1}$.

In particular,

$$\boxed{\Pr(Y = uT/2) = \frac{1}{I_0(\lambda T)}.}$$

For $K = 0$ the unique switch occurs at $T/2$, and $Y = uT/2$ with right-continuous velocity $-u$. For $k \ge 1$, all simplex coordinates are strictly positive almost surely. Attaining $Y = uT/2$ would use all positive duration before the midpoint, leaving extra positive intervals with zero length. Attaining $Y = -uT/2$ would require zero initial positive duration. Both are simplex boundary events. The midpoint lies in an interval with index $1 \le j \le 2k$. For odd j its position is $u(2\sum_{i<j,\ i \text{ even}} d_i - T/2)$; for even j it is $u(T/2 - 2\sum_{i<j,\ i \text{ odd}} d_i)$. Each sum is a nonempty proper prefix of its simplex vector, so on each such region Y is a nonconstant affine function of the free simplex coordinates. Thus the $k \ge 1$ contribution is absolutely continuous on $(-uT/2, uT/2)$.

The velocity convention matters at the atom. On the one-switch component, conditioning instead on $|X_T| < \epsilon$ makes the switch time uniform in a symmetric interval around $T/2$. Half of these paths still have velocity $+u$ at the midpoint and half have $-u$. As $\epsilon \downarrow 0$, their position paths converge uniformly to the one-switch return path, but their midpoint velocity laws retain that equal mixture. Evaluation of velocity at a jump is discontinuous. Our exact bridge uses the right-continuous path version above; an endpoint-window measurement is a separate specified protocol.

4. Cuts sample one preparation

Every finite set of cuts is evaluated on this same count/simplex probability space. If ρ refines π, deleting the extra coordinates of $(X_t, V_t)_{t\in\rho}$ returns $(X_t, V_t)_{t\in\pi}$ pointwise. Hence their joint laws satisfy exact restriction consistency. This argument handles atoms without multiplying singular transition densities.

An independent midpoint-reset rule fails a direct test. Conditional on $Y = uT/2$, speed at most u forces velocity $+u$ almost everywhere on the first half and $-u$ almost everywhere on the second half. In particular $X_{T/4} = X_{3T/4} = uT/4$ deterministically. Giving either quarter point fresh nonzero variance changes the old preparation or violates its speed support. Velocity and the endpoint conditioning carry the required memory.

5. Exact action limit under arbitrary cuts

For any partition π of $[0, T]$, let X^π be the sampled linear interpolant, $|\pi|$ its largest interval, and $S_\pi = S[X^\pi]$. Then

$$S[X] = \frac{mu^2 T}{2}, \qquad 0 \le S[X] - S_\pi \le \frac{mu^2}{2} N_T |\pi|.$$

Indeed the polygon velocity on interval I is the average $\bar{V}_I$, and

$$S[X] - S_\pi = \frac{m}{2} \sum_{I \in \pi} \int_I (V_t - \bar{V}_I)^2 dt.$$

An interval without an interior switch contributes zero. Every other interval contributes at most $mu^2|I|/2$; their total length is at most $N_T|\pi|$. The count is finite almost surely. Differentiating the convergent positive series for I_0, with $I_1 = I_0'$, also gives

$$\mathbb{E}[N_T \mid \text{bridge}] = 1 + z\frac{I_1(z)}{I_0(z)}.$$

Consequently $S_\pi \to mu^2T/2$ almost surely and in L^1 on any deterministic shrinking meshes, with the displayed expected error bound. Also $\|X^\pi - X\|_\infty \leq 2u|\pi|$. Along nested partitions S_π increases, by square completion at each inserted node.

The surviving action relative to the zero-position chord is the kinetic cost of the fixed returning path. The polygon approximation error tends to zero. Its value $mu^2T/2$ tends to zero with duration or speed. These are separate limits from A01's long-observation stationary coefficient $H_* = mu^2/\lambda$. For the bridge midpoint observable, C031 gives $0 \leq \kappa_{\text{mid}} = 4m \operatorname{Var}(Y)/T \leq mu^2T$. Its mean is generally biased, so midpoint action uses $2m\mathbb{E}[Y^2]/T$, rather than variance alone.

6. Consequence for the action-gap programme

Finite speed, velocity memory and exact cut consistency coexist with vanishing polygon error and arbitrarily small action scales. Their combination provides a concrete classical admissible family against which a proposed selection principle can be tested. A positive universal remainder requires an additional restriction on this family, traced to its physical role.

The next test is force control: replace impulses by bounded-acceleration turns and determine the minimum duration of a return with prescribed opposite endpoint velocities. This introduces an energy/force timescale through a mechanical constraint, making its action dependence and scaling testable.

Claims and source coverage are maintained separately in the ledger and the B14 audit. The present path law is a specialization of established telegraph probability; the cut/action consequences are derived here for the project test.

A shared action coefficient from classical composition

A nonnegative mass-dependent fluctuation coefficient becomes mass independent when independent composition preserves its value as a whole-body observable. A single finite-rate, nonzero-speed reference constituent then makes the shared coefficient strictly positive. This conditional result separates the composition premise from the physical clock that sets the value.

1. Observable and independent composition

All masses are positive, positions are on the line, and coefficients have action units. For centered independent displacement variables over a common window $\Delta > 0$, define $\mathrm{Var}(\Delta X_i) = \kappa_i \Delta / m_i$. Let $M = m_1 + m_2$, $\mu = m_1 m_2 / M$, $R = (m_1 X_1 + m_2 X_2)/M$ and $r = X_1 - X_2$. Variance addition gives

$$\kappa_{\mathrm{cm}} := \frac{M}{\Delta}\,\mathrm{Var}(\Delta R) = \frac{m_1\kappa_1 + m_2\kappa_2}{M}, \qquad \kappa_{\mathrm{rel}} := \frac{\mu}{\Delta}\,\mathrm{Var}(\Delta r) = \frac{m_2\kappa_1 + m_1\kappa_2}{M}.$$

Thus $\kappa_{\mathrm{cm}} + \kappa_{\mathrm{rel}} = \kappa_1 + \kappa_2$ and $\mathrm{Cov}(\Delta R, \Delta r) = \Delta(\kappa_1 - \kappa_2)/M$. For independent Gaussian bridges replace Δ by their common covariance kernel; the same coefficient transformation holds. In that Gaussian setting, centre/relative process independence is equivalent to $\kappa_1 = \kappa_2$.

For independent stationary finite irreducible reversible chains J^i, take centered velocities $v_i(J^i)$ and $X_i(t) = \int_0^t v_i(J^i_s)ds$. C019 defines $H_i = 2m_i \int_0^\infty C_i(s)ds$. Independence gives $C_{\mathrm{cm}} = (m_1^2 C_1 + m_2^2 C_2)/M^2$, so the same formulas hold with H in place of κ in the long-window limit. The product state has generator $Q_1 \otimes I + I \otimes Q_2$ and invariant law $\pi_1 \otimes \pi_2$; detailed balance and irreducibility hold factorwise. Retain this state even if distinct state pairs share one velocity value: the projected velocity alone need not be Markov. If each constituent speed is at most $u < c$, the centre speed has the same bound. Relative velocity may reach $2u$; it is a coordinate difference, not a separate particle speed premise.

2. Mass-universality theorem

Assumptions. Fix a preparation class. Every mass $m \in (0, \infty)$ has a finite coefficient $\kappa(m) \geq 0$ depending only on its mass within this class. Independent constituents can be composed, and the coefficient assigned to their whole, of mass M, equals the centre coefficient calculated above. Choose either the common-window coefficient, a Gaussian coefficient, or the Green–Kubo plateau throughout; the theorem does not exchange these observables.

Conclusion. There is a constant $K \geq 0$ such that $\kappa(m) = K$ for every $m > 0$.

Proof. Set $f(m) = m\kappa(m)$. Composition says $f(a+b) = f(a) + f(b)$ on positive reals. For $b > a$, nonnegativity gives $f(b) - f(a) = f(b-a) \geq 0$. Fix a mass unit $m_0 > 0$. Additivity yields $f(qm_0) = qf(m_0)$ for every positive rational q. For any $m > 0$, choose rationals $q_n \uparrow m/m_0$ and $r_n \downarrow m/m_0$. Monotonicity sandwiches $f(m)$ between $q_n f(m_0)$ and $r_n f(m_0)$. Hence $f(m) = mf(m_0)/m_0$, and $K = f(m_0)/m_0$. This also covers $K = 0$.

The premise equating independently composed and whole-body preparations does the physical work. All-positive-mass admissibility makes the rational argument available; a restricted species list needs a separately stated domain theorem.

3. Positive reference theorem

Apply the preceding theorem to Green–Kubo plateaus. Suppose the class contains one reference constituent of mass m_0, stationary symmetric velocities $\pm u_0$, with $0 < u_0 < c$ and reversal rate $0 < \lambda_0 < \infty$. C020 gives

$$\boxed{K = H(m_0) = \frac{m_0 u_0^2}{\lambda_0} > 0, \qquad H(m) = K \quad (m > 0).}$$

This excludes an attained zero in that preparation class using classical probability, nonzero motion, a finite positive clock and the composition premise. The value comes from the reference constituent. Other two-state members consequently obey $\lambda_m = mu_m^2/K$.

For any chosen $K > 0$ and common $0 < u < c$, take independent two-state members with $\lambda_m = mu^2/K$ and include all their finite products. Every product has the same plateau when normalized by its total mass. This realizes coefficient closure even though its centre velocity generally has more than two values. It is a consistency example for the assumptions, not a derivation of the assigned rate law. Across such examples K can approach zero while each individual rate remains finite. A class-wide numerical lower bound needs uniform controls, for example $u_0 \geq u_{\min} > 0$ and $\lambda_0 \leq \Lambda < \infty$ at fixed m_0.

4. Preparation and correlation tests

An environmental label θ allows $\kappa(m,\theta) = K(\theta)$ under the same composition proof at each fixed θ. Universality across such classes requires a preparation-equivalence premise. If cross covariance is $C_{12} = \mathrm{Cov}(\Delta X_1, \Delta X_2)$, the centre coefficient has an additional $2m_1m_2C_{12}/(M\Delta)$; independence fixes it to zero. Bound bodies and common reservoirs must be tested with their actual correlations.

In A02, separately prepared tracers in otherwise identical refreshed baths have plateau $H(m) = (m + M_b)s^2/\nu$. Treating their total mass as one tracer changes the coefficient by

$$H(m_a + m_b) - \frac{m_aH(m_a) + m_bH(m_b)}{m_a + m_b} = \frac{2m_am_bs^2}{\nu(m_a + m_b)}.$$

The positive discrepancy for $s > 0$ identifies failure of the preparation equivalence premise, while preserving the earlier bath calculation. The comparison is between these specified experiments; dynamics of an actual bound composite require their own mechanical model.

5. Scope, source chain and next test

The theorem establishes a conditional universal positive **long-window coefficient**, with action units and a specified preparation class. Its role as a quantum phase constant and its physical-time attraction are subsequent obligations. At bounded speed the microscopic variance coefficient still tends to zero by C018. The next A09 calculation connects these separate windows to the real telegraph and complex checkerboard propagators.

This note develops the user's composition suggestion and the independent P02 product-chain calculation. C019–C020 supply the reviewed correlation input; Pavliotis (2010) gives its Poisson-equation source, and Pitman–Yor (2018) the Gaussian covariance background. Per-result prior-art status belongs to B16. The five dedicated algebra checks cover the two coefficients, their sum, their cross covariance and the bath discrepancy. Existing P02 checks also test a nine-state product generator. The written proofs supply the general additive-function and positive-reference conclusions.

Exact midpoint crossover for a finite-speed return bridge

The velocity-resolved return bridge has an explicit midpoint variance:

$$\frac{\kappa_{\mathrm{mid}}(T)}{H_*} = g(z) = zR(z)[1 - R(z)], \qquad R(z) = \frac{\int_0^z I_0(s)\,ds}{zI_0(z)}, \qquad z = \lambda T.$$

It has cubic onset $g(z) = z^3/6 + O(z^5)$ and converges to one as $z \to \infty$. Thus the conditioned observable reaches the same long-window plateau as the stationary increment observable, through a different crossover function.

1. Preparation and units

Fix $m > 0$, $0 < u < c$, $\lambda > 0$ and $T > 0$. Use the C033 bridge from $(X_0, V_0) = (0, +u)$ to $(X_T, V_T) = (0, -u)$, with its explicit density disintegration and right-continuous velocity convention. Put $Y = X_{T/2}$, $Q = 2Y/(uT)$ and $N_T = 2K + 1$. Its count weights are

$$w_k = \frac{(z/2)^{2k}}{(k!)^2 I_0(z)}, \qquad k = 0, 1, \ldots.$$

Here $I_0(z) = \sum_{k\geq 0}(z/2)^{2k}/(k!)^2$. The action-valued observables are $\kappa_{\mathrm{mid}} = 4m\,\mathrm{Var}(Y)/T$ and $A_2 = 2m\mathbb{E}[Y^2]/T$, while $H_* = mu^2/\lambda$ is the C020 stationary long-window plateau. All expectations below use this bridge; cut refinement keeps this preparation fixed and does not vary T.

2. A beta law at fixed switch count

For $k = 0$, $Q = 1$ deterministically. For every integer $k \geq 1$,

$$\boxed{\left.\frac{Q+1}{2}\,\right|\, K = k \;\sim \mathrm{Beta}(k+1, k).}$$

Equivalently its density on $-1 < q < 1$ is

$$f_k(q) = \frac{(2k-1)!}{2^{2k-1}[(k-1)!]^2}(1+q)(1-q^2)^{k-1}.$$

Proof. For a telegraph segment of duration t, displacement x and initial velocity $+u$, let $a = (t + x/u)/2$, $b = (t - x/u)/2$. Uniform Poisson spacings, grouped into positive and negative durations, give joint position/count densities in the interior $|x| < ut$:

$$d_{2j+1}(x,t) = \frac{e^{-\lambda t}\lambda^{2j+1}}{2u(j!)^2}a^j b^j \quad (j \geq 0),$$

$$d_{2j}(x,t) = \frac{e^{-\lambda t}\lambda^{2j}}{2u\,j!(j-1)!}a^j b^{j-1} \quad (j \geq 1).$$

The zero-count law is an atom at ut, handled separately. For initial velocity $-u$, reverse the displacement sign. These formulas follow from Dirichlet spacings with respectively $(j+1, j+1)$ and $(j+1, j)$ groups; the position Jacobian is $1/(2u)$.

Split the return path at $t = T/2$ and take an interior midpoint y. The two possible midpoint velocity sectors have even/odd and odd/even counts. With $a = T(1+q)/4$, $b = T(1-q)/4$, each sector contributes $a^k b^{k-1}$ times a factorial sum. Their combined joint density in midpoint y and terminal position zero is

$$\frac{e^{-\lambda T}\lambda^{2k+1}}{2u^2}a^k b^{k-1} S_k, \qquad S_k = \sum_{j=1}^{k} \frac{1}{j!(j-1)![(k-j)!]^2} = \frac{k(2k)!}{2(k!)^4}.$$

For the last equality write each term as $j\binom{k}{j}^2/(k!)^2$, use symmetry $j \leftrightarrow k-j$ and Vandermonde's identity $\sum_j \binom{k}{j}^2 = \binom{2k}{k}$. Divide by C033's terminal position/count density $e^{-\lambda T}\lambda^{2k+1}(T/2)^{2k}/[2u(k!)^2]$ and multiply by $dy/dq = uT/2$. This yields f_k. It integrates to one, so at $k \geq 1$ the interior formula exhausts the conditional mass. At $k = 0$ retain the deterministic midpoint atom; its mixture mass is $1/I_0(z)$.

3. Exact moments and limits

Beta integration gives, also including the $k = 0$ atom,

$$\mathbb{E}[Q \mid K = k] = \mathbb{E}[Q^2 \mid K = k] = \frac{1}{2k+1}.$$

Summing the absolutely convergent series and integrating I_0 term by term therefore gives $\mathbb{E}Q = \mathbb{E}Q^2 = R(z)$. Consequently

$$\mathbb{E}Y = \frac{uT}{2}R(z), \qquad \mathrm{Var}(Y) = \frac{u^2T^2}{4}R(z)[1-R(z)],$$
$$\kappa_{\mathrm{mid}} = H_* g(z), \qquad A_2 = \frac{H_*}{2}zR(z).$$

For $z > 0$, $0 < R(z) < 1$, hence $g(z) > 0$. The exact identity also improves the generic midpoint bound to $g(z) \leq z/4$. The Taylor series gives $R(z) = 1 - z^2/6 + O(z^4)$ and thus the cubic onset in the opening result.

For the large-window limit use the integral representation $I_0(z) = \pi^{-1}\int_0^\pi e^{z\cos\theta}d\theta$, DLMF 10.32.1. The normalized exponential weight concentrates at $\theta = 0$: for any $0 < \delta < \pi$, the weight of $[\delta, \pi]$ is at most $(2\pi/\delta)e^{-z(\cos(\delta/2)-\cos\delta)}$. Differentiation under the integral then gives $I_0'(z)/I_0(z) \to 1$. Both $I_0(z)$ and $\int_0^z I_0(s)ds$ diverge, so l'Hopital's rule yields

$$zR(z) = \frac{\int_0^z I_0(s)ds}{I_0(z)} \longrightarrow 1, \qquad R(z) \longrightarrow 0, \qquad g(z) \longrightarrow 1.$$

At fixed m, u, λ, this proves

$$\kappa_{\mathrm{mid}}(T) \to H_*, \qquad A_2(T) \to H_*/2, \qquad \mathbb{E}Y \to \frac{u}{2\lambda} \quad (T \to \infty).$$

The normalized mean $\mathbb{E}Q$ vanishes; the unscaled midpoint mean retains a boundary bias. P02's sampler anticipated the plateau and cubic onset. This proof supplies the full curve and corrects its provisional statement that the unscaled midpoint mean tends to zero. Monotonicity of g is a separate question; the endpoint limits do not assume it.

4. Crossover windows and mass universality

For either C018's increment coefficient or C031's return-midpoint coefficient, a prescribed value $K_* > 0$ under a speed ceiling u can be realized only if

$$T \geq T_* := \frac{K_*}{mu^2}.$$

This follows by dividing $K_* \leq mu^2T$ by the positive factor mu^2. For the present bridge the sharper $g \leq z/4$ gives $\kappa_{\mathrm{mid}} \leq mu^2T/4$. These are necessary bounds on the observable, not sufficient conditions for any prescribed distribution.

A consequence for C035 is immediate. Suppose masses extend arbitrarily close to zero, the speed ceiling u and finite window T are common, and the increment or midpoint coefficient is the same $K_* \geq 0$ for every mass. Then $K_* \leq mu^2T$ for all those masses forces $K_* = 0$. A positive mass-universal plateau is compatible with finite speed through mass-dependent crossover windows, rather than a common finite window valid for arbitrarily small masses.

For a telegraph family with shared plateau H_* and common speed u, $\lambda_m = mu^2/H_*$ and $T_{*,m} = 1/\lambda_m$ when $K_* = H_*$. The exact midpoint curve is $H_* g(mu^2 T/H_*)$. Its dependence on mT exhibits the nonuniformity of the large-window limit across small masses. The stationary increment instead uses $H_*[1 - (1 - e^{-2\lambda_m T})/(2\lambda_m T)]$.

At the additional formal identifications $H_* = \hbar$ and $u = c$, the scales are $T_{*,m} = \hbar/(mc^2)$ and $uT_{*,m} = \hbar/(mc)$, the reduced Compton time and length. The bounded-speed inequalities extend to $u = c$; the identification with a quantum phase scale belongs to the next dynamics test. This calculation concerns classical probability and specified windows.

5. Research handoff

A09a supplies the exact conditioned crossover and the common-window test. A09b retains the direct checkerboard source reading, complex-amplitude recurrence and quantum normalization. B17 records the per-result source audit; Cinque's occupation-time formulas provide the telegraph source chain, while the beta midpoint specialization and action normalization are derived here. The existing C033 path law remains unchanged. Keep physical time, observation duration and the cut mesh separate when applying these results to the target.

One action scale, two rules for composing paths

The same inverse-time parameter can generate a classical telegraph process or a unitary Dirac evolution. The distinction lies in the rule for composing path weights. We give an exactly normalized checkerboard step, its strong wavepacket limit, and a measurement test that separates its corner amplitude from a classical reversal probability. This completes A09b's mathematical comparison and identifies the remaining physical selection question.

1. Objects and physical premises

Fix a mass $m > 0$, propagation speed $c > 0$, and action parameter $K > 0$. Set $\omega = mc^2/K$, with units of inverse time. A time step $\varepsilon > 0$ has spatial length $c\varepsilon$. Components labelled $+$ and $-$ move right and left. The classical comparison uses the same speed as a model parameter; identifying it with the physical speed of light is an additional specialization.

Use Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad (S_\varepsilon f)_\pm(x) = f_\pm(x \mp c\varepsilon).$$

Classical states are nonnegative two-component probability densities or measures, with total mass one. Coherent states belong to $\mathcal{H} = L^2(\mathbb{R}; \mathbb{C}^2)$, with norm one and Born probabilities. The common translation space makes a strong refinement limit well-defined. At each fixed mesh the same formulas also define a walk on $\ell^2(c\varepsilon\mathbb{Z}; \mathbb{C}^2)$.

C035–C036 supply a candidate positive classical plateau within a specified composition/preparation class. Here K is supplied, and equating it to that plateau is a comparison hypothesis. Complex linear superposition, the squared norm measurement rule, and the chosen mass-mixing generator specify the coherent model. The calculation locates their use rather than deriving them from classical composition.

2. Classical transitions and coherent corners

The classical step is $S_\varepsilon P_\varepsilon$, where

$$P_\varepsilon = (1 - \lambda\varepsilon)I + \lambda\varepsilon\sigma_x, \qquad 0 \le \lambda\varepsilon \le 1.$$

Its entries are probabilities and its columns sum to one. On smooth densities its first-order generator is $A_\lambda = -c\sigma_z\partial_x + \lambda(\sigma_x - I)$. Thus the continuum component equations describe the classical telegraph process with flip rate λ. Their sum satisfies $\rho_{tt} + 2\lambda\rho_t = c^2\rho_{xx}$.

The coherent step instead is

$$C_\varepsilon = \frac{I - i\omega\varepsilon\sigma_x}{\sqrt{1 + \omega^2\varepsilon^2}}, \qquad U_\varepsilon = S_\varepsilon C_\varepsilon.$$

Exact normalization. Since $\sigma_x^2 = I$ and σ_x is Hermitian, $C_\varepsilon^\dagger C_\varepsilon = I$. Translation is unitary, so every U_ε preserves the wavepacket norm. Expanding a product of n steps gives a path factor $(1 + \omega^2\varepsilon^2)^{-n/2}(-i\omega\varepsilon)^r$ for r flips, counting possible mixing before the first translation.

The fixed-first-step convention of Skopenkov–Ustinov, Definition 2, instead has $n - 1$ mixing opportunities and an initial overall phase i. Its source components (a_1, a_2) obey Proposition 5. The explicit change of basis

$$\psi = W\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \qquad W = \begin{pmatrix} 0 & 1 \\ -i & 0 \end{pmatrix}$$

turns that recurrence into U_ε in natural units, after replacing the source's $m\varepsilon$ by $\omega\varepsilon$ and its spatial step by $c\varepsilon$. Indeed W is unitary, swaps the two translation directions, and takes the source coin $\left(\begin{smallmatrix}1 & r\\ -r & 1\end{smallmatrix}\right)/\sqrt{1+r^2}$ to $(I - ir\sigma_x)/\sqrt{1+r^2}$. This is an established checkerboard construction in an explicitly matched basis; B18 records the prior-art audit.

3. Strong continuum limit (C039)

For every fixed $T \geq 0$ and $\psi \in \mathcal{H}$,

$$U_{T/N}^N \psi \longrightarrow e^{-iTH_D/K}\psi \quad \text{in } \mathcal{H}, \qquad H_D = -iKc\sigma_z\partial_x + mc^2\sigma_x.$$

Proof. Use the unitary Fourier transform with phase $e^{-ipx/K}$; p is momentum. The step multiplier is

$$M_\varepsilon(p) = e^{-icp\varepsilon\sigma_z/K}\frac{I - i\omega\varepsilon\sigma_x}{\sqrt{1+\omega^2\varepsilon^2}} = I - \frac{i\varepsilon}{K}(cp\sigma_z + mc^2\sigma_x) + O_p(\varepsilon^2).$$

At fixed p, compare M_ε with $e^{-i\varepsilon H(p)/K}$, where $H(p) = cp\sigma_z + mc^2\sigma_x$. Both matrices are unitary. A telescoping sum bounds the N-step difference by $NO_p((T/N)^2)$, which tends to zero. Its norm is also bounded by two for all p, N. Dominated convergence against $|\widehat{\psi}(p)|^2$ proves the strong limit. The Hermitian multiplier $H(p)$ defines a self-adjoint operator with domain $H^1(\mathbb{R};\mathbb{C}^2)$, since

$$H(p)^2 = (c^2p^2 + m^2c^4)I.$$

Consequently the limiting evolution is unitary and each component solves $\partial_t^2\psi - c^2\partial_x^2\psi + \omega^2\psi = 0$ distributionally. The spectrum has branches $\pm E_p$, where $E_p = \sqrt{c^2p^2 + m^2c^4}$. Their rest separation is $2mc^2$ in energy units. This is a single-particle branch separation; the vacuum spectral problem belongs to a field-theory construction.

This proof concerns fixed L^2 initial data. The source's light-cone delta-function warning concerns the singular point-source kernel and its undefined square. It leaves the wavepacket norm identity above intact.

4. Analytic continuation and the nonrelativistic limit

At the differential-equation level, put $\lambda = -i\omega$ and remove the scalar phase: $f(t) = e^{i\omega t}\psi(t)$. Then $f_t = A_{-i\omega}f$ gives $\psi_t = -c\sigma_z\psi_x - i\omega\sigma_x\psi$. The opposite continuation sign gives the unitarily equivalent opposite mass sign. At finite mesh the continued stochastic coin is

$$\widetilde{P}_\varepsilon = (1 + i\omega\varepsilon)I - i\omega\varepsilon\sigma_x.$$

Its symmetric eigenvector has squared norm factor one, and its antisymmetric eigenvector has factor $1+4\omega^2\varepsilon^2$. No scalar phase or common normalization makes both factors one when $\omega\varepsilon \neq 0$. Our normalized coherent coin has the same generator after phase removal, but differs at second order. This identifies precisely the continuum sense of the analytic-continuation correspondence.

For the positive branch, subtract the rest energy before taking $c \to \infty$ at fixed m, K, p:

$$E_p - mc^2 = \frac{p^2}{m(\sqrt{1 + p^2/(m^2c^2)} + 1)} \longrightarrow \frac{p^2}{2m}.$$

In the scalar spectral representation of that branch, dominated convergence of the unit-modulus multipliers gives strong convergence on $L^2(\mathbb{R})$ to the free Schrödinger group with action scale K. This statement uses branch selection and rest-phase removal, and takes the mesh limit first. It is distinct from the source's point-source, large-time triple limit.

For the real process, $\lambda = mc^2/K$ instead gives Kac's diffusion scaling $c^2/(2\lambda) = K/(2m)$, as in C021: its limit is the heat process. Equal dimensional coefficients organize the comparison; the path-weight rule decides which evolution results.

5. Observing every corner changes the limit (C040)

Perform an ideal projective direction measurement after each coherent step, starting from a wavepacket with definite direction. Given the previous direction, the next reversal has probability

$$q_\varepsilon = \frac{\omega^2\varepsilon^2}{1+\omega^2\varepsilon^2}.$$

Proof and limit. The off-diagonal coin amplitude is $-i\omega\varepsilon/\sqrt{1+\omega^2\varepsilon^2}$; translations preserve its component norm. Projection resets direction after every step. Hence the conditional flip probability is q_ε at each step, independently of the preceding spatial profile. For $N = T/\varepsilon$,

$$\Pr(\text{at least one flip}) \le Nq_\varepsilon \le \omega^2 T\varepsilon \longrightarrow 0.$$

With probability tending to one the initially selected direction persists for the whole interval. The observed walk has a ballistic fixed-ω limit. The classical telegraph step instead has flip probability $\lambda\varepsilon$ and a finite mean flip count λT. Thus $\lambda = \omega$ equates two coefficients with inverse-time units, not their trajectory or measurement laws.

6. Next physical test

The classical positive plateau and the coherent action parameter now have an exact algebraic comparison. The remaining obligation is an operational rule for coherent composition that identifies their values across preparations. Inserted observation cuts must be distinguished from actual repeated measurements: Section 5 gives different limiting laws for these operations. G01 next tests which velocity modes the plateau controls and whether that control supplies a spectral lower bound.

Sources and reproduction

Skopenkov and Ustinov, *Feynman checkers: towards algorithmic quantum theory.*

Selected arXiv version, 28 February 2022: Definition 2, Propositions 5–6, pp. 12–13; proof p. 31; singular-kernel discussion p. 19. The source companion records exact coverage and the older-source page tasks. C039 is a reconstruction with an elementary wavepacket proof; C040 is a derived protocol consequence.

Run `python3 scripts/checkerboard_checks.py` for algebra, finite paths and Fourier-mode convergence checks. The general limits rest on the proofs above.

When an action plateau controls a spectral gap

A complete collection of observables with bounded total susceptibility gives a lower relaxation-gap bound. A single positive velocity plateau instead gives an upper bound, and can miss an arbitrarily slow internal mode. We prove both statements for finite reversible dynamics and specify the energy normalization needed for the Dirac comparison.

1. Operator, observable and units

Let Q generate an irreducible continuous-time Markov chain on a finite set of $n \geq 2$ states, with stationary probabilities $\pi_i > 0$ and detailed balance $\pi_i Q_{ij} = \pi_j Q_{ji}$. The generator acts on functions, with row sums zero. On $L^2(\pi)$ use $\langle f, g\rangle_\pi = \sum_i \pi_i \overline{f_i} g_i$. The centered space $\mathcal{H}_0 = \{f : \langle 1, f\rangle_\pi = 0\}$ has dimension $n-1$. All domains are the entire indicated finite-dimensional space.

Set $A = -Q|_{\mathcal{H}_0}$. It is positive self-adjoint with eigenvalues $0 < \gamma_1 \leq \cdots \leq \gamma_{n-1}$. The full relaxation gap is γ_1, in inverse-time units. For a real centered observable v with velocity units, define

$$\chi(v) = \langle v, A^{-1}v\rangle_\pi = \int_0^\infty \langle v, e^{tQ}v\rangle_\pi \, dt, \qquad H(v) = 2m\chi(v), \quad m > 0.$$

χ has units length squared per time, and H has action units. This is C019's long-observation plateau for the integrated velocity. The Green–Kubo/Poisson representation is matched to Pavliotis in B20; the finite-dimensional consequences below are derived explicitly. Time here is the chain's evolution parameter.

2. One observable: the product and its direction (C041)

Write $v = \sum_j a_j e_j$ in an orthonormal eigenbasis of A, and $\sigma_v^2 = \|v\|_\pi^2 > 0$. Then

$$H(v) = 2m \sum_j \frac{|a_j|^2}{\gamma_j}, \qquad H(v)\gamma_1 \leq 2m\sigma_v^2 \leq H(v)\gamma_{n-1}.$$

Proof. Integrate each exponential in $\langle v, e^{tQ}v\rangle_\pi = \sum_j |a_j|^2 e^{-\gamma_j t}$ and use $\sum_j |a_j|^2 = \sigma_v^2$. The left inequality gives $\gamma_1 \leq 2m\sigma_v^2/H(v)$: an upper gap bound. The normalized integral time $\tau_v = \chi(v)/\sigma_v^2$ is a weighted average of $1/\gamma_j$; in particular $\tau_v \leq 1/\gamma_1$.

For the two-state chain $Q = \lambda(\sigma_x - I)$ and $v = u(1, -1)$, $\pi = (1/2, 1/2)$, with $u, \lambda > 0$, the centered space is one-dimensional. Therefore

$$\gamma_1 = 2\lambda, \qquad H(v) = \frac{mu^2}{\lambda}, \qquad H(v)\gamma_1 = 2mu^2.$$

Here σ_x exchanges the two signs. This equality identifies the source of the two-state product: every centered mode is visible to velocity.

3. Fixed plateau with a closing full gap (C041)

Take two independent signs $s, r \in \{-1, 1\}$ with rates $\lambda > 0$ and $\epsilon > 0$, respectively, and define velocity $v(s, r) = us$, with fixed $0 < u < c$. The four-state generator is

$$Q_{\lambda,\epsilon} = \lambda(\sigma_x - I) \otimes I + I \otimes \epsilon(\sigma_x - I).$$

The uniform measure is stationary and reversible. The functions $1, s, r, sr$ form an orthonormal eigenbasis with eigenvalues of $-Q$ equal to $0, 2\lambda, 2\epsilon, 2(\lambda + \epsilon)$. Velocity overlaps only the s mode. Thus

$$H(v) = \frac{mu^2}{\lambda}, \qquad \gamma_1 = 2\min(\lambda, \epsilon) \longrightarrow 0 \quad \text{as } \epsilon \downarrow 0.$$

Every positive-ϵ member is irreducible, with the same bounded speed and the same positive plateau. At the limiting parameter the label becomes conserved. The velocity path law itself is unchanged throughout the family. The missing information is the label's relaxation: finite-rate assumptions for each member give positivity member by member, while a uniform lower gap requires additional control over the family.

4. A sufficient condition: complete observability (C042)

Choose centered velocity-unit observables $f_1, \ldots, f_r$ and suppose that, for every $g \in \mathcal{H}_0$,

$$\sum_{a=1}^{r} |\langle f_a, g\rangle_\pi|^2 \geq \alpha \|g\|_\pi^2, \qquad \alpha > 0.$$

This *frame bound* means that the collection detects every centered mode; α has velocity-squared units. Put $S = \sum_a \chi(f_a)$. Then

$$\boxed{\gamma_1 \geq \frac{\alpha}{S}.}$$

If $H(f_a) \leq B_a$, it follows that $\gamma_1 \geq 2m\alpha / \sum_a B_a$.

Proof. Set $b_j = \sum_a |\langle e_j, f_a\rangle_\pi|^2$. The frame hypothesis gives $b_j \geq \alpha$ for every unit eigenvector. Hence

$$S = \sum_j \frac{b_j}{\gamma_j} \geq \frac{\alpha}{\gamma_1}.$$

Multiplication by γ_1/S proves the result. Equivalently, the frame operator $F = \sum_a |f_a\rangle\langle f_a|$ satisfies $F \geq \alpha I$ and $S = \mathrm{Tr}(A^{-1}F)$.

For a family, uniform $\alpha \geq \alpha_0 > 0$ and $S \leq S_0 < \infty$ give the uniform bound $\gamma_1 \geq \alpha_0/S_0$. When masses vary, the action-valued version must control $\sum_a B_a/(2m)$, rather than merely $\sum_a B_a$. The frame can have full rank only if $r \geq n-1$. These requirements expose the cost of carrying the estimate to larger systems.

The exact all-observable formulation is

$$\sup_{f \in \mathcal{H}_0 \setminus \{0\}} \frac{\chi(f)}{\|f\|_\pi^2} = \|A^{-1}\| = \frac{1}{\gamma_1}.$$

The spectral expansion proves the upper bound, and a slowest eigenvector attains equality. It is the inverse-operator version of a Poincaré bound.

In the four-state example choose $f_1 = us$, $f_2 = ur$, $f_3 = usr$. They give $F = u^2 I$ on $\mathcal{H}_0$, and

$$S = u^2 \left[\frac{1}{2\lambda} + \frac{1}{2\epsilon} + \frac{1}{2(\lambda+\epsilon)}\right].$$

The complete collection detects the approaching gap closure through $\chi(f_2) \to \infty$. Each observable stays bounded in magnitude by u. Thus speed control and susceptibility control address different premises.

5. Energy normalization and the companion field problem

On full $L^2(\pi)$ define $\mathcal{E} = K(-Q)$ with a supplied action unit $K > 0$. Constants form its unique zero-energy ground space; its excitation gap is $\Delta_{\mathcal{E}} = K\gamma_1$. In the two-state model, setting $K = H(v)$ gives

$$\Delta_{\mathcal{E}} = 2mu^2.$$

The numerical substitution $u = c$ matches C039's Dirac rest-branch separation $2mc^2$. The two operators have different state spaces and spectral meanings: $\mathcal{E}$ is a nonnegative finite-state relaxation operator, while the Dirac operator has positive and negative single-particle branches. A quantum field vacuum gap additionally requires its physical Hilbert space and the continuum/infinite-volume construction.

For the companion programme, C042 supplies a candidate form of estimate: observable coverage bounded below, inverse-generator response bounded above. Transferring it requires identifying the physical generator and proving that both bounds survive the relevant limits. An algorithmic sampling chain's relaxation time is a distinct object from physical Euclidean time evolution.

6. Next test and reproduction

G02 asks whether mechanical composition preserves a useful observable collection and its quantitative coverage. Start with the four-state example, then weaken access to the label. Track the smallest frame eigenvalue and total susceptibility together; a gap bound based on either one alone loses the information isolated above. A09 continues the independent action-scale and coherent-composition selection task.

Run `python3 scripts/susceptibility_gap_checks.py`. The B20 companion records the Green–Kubo source match and the bounded prior-art coverage. C041 is a spectral/product-chain consequence of C019; C042 is an elementary finite-dimensional observability criterion, with no novelty claim.

A sharp mechanical cost for a finite-duration reversal

A return with prescribed endpoint velocities $+u$ and $-u$ and acceleration ceiling a requires duration at least $2u/a$. Its minimum kinetic action is $mu^3/(3a)$, independent of any additional waiting time. Meanwhile its sampled polygon action converges with a sharp quadratic mesh bound. The endpoint reversal cost and the refinement error therefore have different limits.

1. Model and sharp minimum (C043)

Fix $m, a, T > 0$ and $0 < u < c$ in a specified inertial frame. The admissible paths are $X \in W^{2,\infty}(0,T)$, with continuous velocity representative $v = \dot{X}$,

$$X(0) = X(T) = 0, \quad v(0) = u, \quad v(T) = -u, \quad |v| \le u, \quad |\dot{v}| \le a \quad \text{almost everywhere.}$$

The functional is the kinetic cost $S_K[X] = (m/2)\int_0^T v^2 dt$, in action units. The acceleration is an admissible external control, with force $m\dot{v}$. This is a bounded-control mechanics problem; a prescribed autonomous potential, its potential contribution to Hamilton's action and a dynamical momentum receiver would specify additional physical data.

Proposition. The class is nonempty exactly when $T \ge 2u/a$. For such T, the unique minimizing velocity is, with $\tau = u/a$,

$$v_*(t) = \begin{cases} u - at, & 0 \le t \le \tau, \\ 0, & \tau \le t \le T - \tau, \\ -a(t - T + \tau), & T - \tau \le t \le T. \end{cases} \qquad S_{K,\min} = \frac{mu^3}{3a}.$$

Proof. Lipschitz continuity gives $2u = |v(T) - v(0)| \le aT$. For a feasible duration the intervals $[0,\tau]$ and $[T-\tau, T]$ have disjoint interiors. On the first, $v(t) \ge u - at \ge 0$; on the last, $v(t) \le -u + a(T-t) \le 0$. Hence

$$\int_0^T v^2 dt \ge \int_0^\tau (u - at)^2 dt + \int_{T-\tau}^T [u - a(T-t)]^2 dt = \frac{2u^3}{3a}.$$

The displayed v_* has zero total integral and therefore returns to the starting position. It obeys both ceilings and attains equality. Equality forces each endpoint ramp and zero velocity in the middle almost everywhere; continuity gives uniqueness. Integration with $X(0) = 0$ fixes the path. The maximum excursion is $u^2/(2a)$, reached at the waiting segment.

At $T = 2u/a$ the velocity is $u - at$ throughout. Indeed equality in the total velocity-change bound forces $\dot{v} = -a$ almost everywhere, so the whole admissible class is then a singleton. At longer durations the admissible class includes variations even though the minimizing path remains unique.

2. Which premises hold the cost above zero?

The bound depends on prescribed nonzero endpoint speed and a finite acceleration ceiling. At fixed force ceiling $F_{\max} = ma$ it becomes

$$S_{K,\min} = \frac{m^2u^3}{3F_{\max}}.$$

For fixed m, a, T, choosing successively smaller positive u eventually preserves feasibility and gives $S_{K,\min} \to 0$. An upper speed ceiling allows this family. Likewise, increasing the permitted acceleration at fixed m, u, T drives the minimum to zero. Uniform positivity over a preparation class requires corresponding lower bounds on mass and endpoint speed and an upper bound on acceleration, or another premise controlling their combination.

For $T > 2u/a$, choose a nonzero smooth function ϕ supported strictly inside the waiting interval, with $\int \phi = 0$. Such functions are obtained by differentiating a nonconstant smooth compactly supported bump. For small $b \neq 0$, $v_b = v_* + b\phi$ satisfies the same speed/acceleration bounds and all endpoint conditions. Since the supports of ϕ and the nonzero part of v_* are disjoint,

$$S_K[v_b] - S_K[v_*] = \frac{mb^2}{2} \int \phi^2 dt \longrightarrow 0.$$

Thus the positive minimum total cost coexists with positive excess costs accumulating at zero. Its source is the endpoint-conditioned motion, rather than a discrete spacing of admissible action values. The cost is frame-specific: the endpoint velocities and return condition already select that frame.

3. The sampled polygon error (C044)

For any partition $\pi : 0 = t_0 < \cdots < t_N = T$, write $d_i = t_{i+1} - t_i$, $|\pi| = \max_i d_i$, and define the chord kinetic action

$$S_\pi = \frac{m}{2} \sum_i \frac{[X(t_{i+1}) - X(t_i)]^2}{d_i}.$$

For every admissible path, and more generally every velocity with Lipschitz constant at most a,

$$0 \leq S_K - S_\pi \leq \frac{ma^2}{24} \sum_i d_i^3 \leq \frac{ma^2 T}{24} |\pi|^2.$$

Proof. The chord velocity on each cell is its average $\bar{v}_i$. Square completion and the pair-variance identity give

$$S_K - S_\pi = \frac{m}{2} \sum_i \int_{t_i}^{t_{i+1}} (v - \bar{v}_i)^2 dt,$$

$$\int_I (v - \bar{v}_I)^2 dt = \frac{1}{2d} \int_I \int_I [v(s) - v(t)]^2 ds\, dt \leq \frac{a^2}{2d} \int_0^d \int_0^d (s-t)^2 ds\, dt = \frac{a^2 d^3}{12}.$$

Summing proves the claim. The constant $1/24$ is sharp: the feasible minimum-duration return has affine velocity with slope $-a$ on every cell, and attains the first upper bound for every partition. Equal-width cells also attain the mesh bound. In the longer-duration minimizer, cells contained inside a ramp saturate the local bound and cells inside the waiting interval have zero error.

At fixed m, a, T this convergence is uniform over the admissible class. For families with increasing acceleration bound a_N, the estimate still forces the defect to vanish if $a_N|\pi_N| \to 0$ (fixed m, T). Maintaining a positive defect along shrinking meshes therefore requires loss of this uniform acceleration control. This is a necessary condition, not a construction of a positive remainder.

4. Source comparison and next mechanical step

Liberzon's double-integrator example provides the bounded-acceleration control framework and a minimum-time problem with running cost one. Here the endpoint data and running cost $mv^2/2$ are different; the endpoint-envelope proof above establishes the exact result. B21 records the bounded prior-art search.

The next mechanical question is to realize finite-duration turns with a specified conservative receiver or interaction, then derive its correlation law. A07's external-control class provides a benchmark for such a realization. In the companion G02 task, the receiver's internal degrees of freedom also test which modes the available observables detect.

Run `python3 scripts/bounded_acceleration_checks.py` for exact integrals, matching conditions, variations and rational partition tests. The written proofs establish the general minima and uniform convergence.